Check for updates

Geological Journal

WILEY

RESEARCH ARTICLE OPEN ACCESS

Geochemistry and Geochronology of Magmatic Rocks in Southwestern and Northeastern Cambodia: Implications for the Tectonic Evolution and Metallogenesis of the Loei Fold Belt

Sirisokha Seang[1,2] | Khin Zaw[3] | Kakda Kret[1,4] | Ivan Belousov[3] | Kotaro Yonezu[5] | Koichiro Watanabe[6] | Charles Makoundi[3]

[1]Faculty of Geo-Resources and Geotechnical Engineering, Institute of Technology of Cambodia, Phnom Penh, Cambodia | [2]Materials Science and Structure Unit, Research and Innovation Center, Institute of Technology of Cambodia, Phnom Penh, Cambodia | [3]CODES Centre of Ore Deposit and Earth Sciences, University of Tasmania, Hobart, Tasmania, Australia | [4]Energy Technology and Management Unit, Research and Innovation Center, Institute of Technology of Cambodia, Phnom Penh, Cambodia | [5]Department of Earth Resources Engineering, Kyushu University, Fukuoka, Japan | [6]Japan International Cooperation Agency, Tokyo, Japan

Correspondence: Charles Makoundi (c.makoundi@utas.edu.au)

Received: 25 October 2025 | **Revised:** 10 May 2026 | **Accepted:** 18 May 2026

Handling Editor: Yongjiang Liu

Keywords: Cambodia | Loei Fold Belt | Paleo-Tethys Ocean | REE | zircon U–Pb geochronology

ABSTRACT

The Phnom Sro Ngam and Halo Prospects are potentially important targets for exploring porphyry copper-molybdenum, epithermal and skarn-related gold deposits in Cambodia. These areas are in the southwest and northeast of Cambodia, extending from the highly mineralized Loei Fold Belt in Thailand and Laos. However, the metallogenic characteristics and tectonic significance of magmatic rocks within this fold belt in Cambodia remain ambiguous. In this paper, we present new geochemical data and zircon U–Pb ages for the magmatic rocks from the Halo and the Phnom Sro Ngam prospects in northern and southwestern Cambodia. Geochemically, the magmatic rocks from both areas in Cambodia were formed and emplaced in a subduction-related tectonic environment under a volcanic arc setting. All magmatic rocks plot into I-type and S-type granite on the ($Na_2O + K_2O$) versus (Zr + Nb + Ce + Y) diagram. Trace element spider diagrams normalized to primitive mantle display a strong enrichment in large-ion lithophile elements such as Rb, Ba and K and depletion in some high-field-strength elements such as Nb and Ti, suggesting that the magma was generated in a subduction-related tectonic setting. The REE patterns show slight enrichments in light REEs relative to heavy REEs and weak negative Eu anomalies. A distinct negative Eu anomaly indicates possible fractionation of Eu to plagioclase. Considering La/Yb as a crustal thickness proxy in intermediate to felsic calc-alkaline rocks, the arc crustal thickness in Phnom Sro Ngam was probably calc-alkaline granitoid. Zircon U–Pb dating yielded 229 to 282 Ma corresponding to Late Permian to Triassic and they are similar to magmatic activity in the Loei Fold Belt, which occurred during the subduction of the Paleo-Tethys Oceanic crust towards the east beneath the Indochina terrane. The trace element geochemistry of the zircon grains strongly indicates great potential in porphyry copper mineralization in the prospects.

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2026 The Author(s). *Geological Journal* published by John Wiley & Sons Ltd.

1 | Introduction

Gold, copper, lead and zinc mineralization in Cambodia is closely associated with widely distributed magmatic rocks in the northeastern, northern and southwestern parts of the country (Geopacific Resources Ltd 2014; Zaw et al. 2014; Angkor Resources Corp 2018; Kasahara et al. 2021; Uchida et al. 2023). The Halo copper-molybdenum Prospect is located in Oyadav, Ratanakiri, Cambodia, approximately 450 km northeast of the capital city of Phnom Penh. The Phnom Sro Ngam area is situated in the Chhouk district of Kampot Province, approximately 120 km from Phnom Penh. Phnom Sro Ngam is a potential target for exploring porphyry, epithermal and skarn-related copper-gold deposits (Samnang Angkor Development 2019; Ly et al. 2022), and the Halo Prospect is a target for porphyry copper-molybdenite deposit (Angkor Resources Corp 2018; Seang et al. 2019).

The Halo and Phnom Sro Ngam Prospects are located in the Loei Fold Belt (LFB) (Zaw et al. 2014; Cheng et al. 2019). However, the previous research on LFB mainly focused on the geochronology and geochemistry of magmatic rocks in Thailand and Laos, and some parts of Cambodia (e.g., Zaw et al. 2014; Qian et al. 2015). Only a few researchers have focused on the extension of the LFB into Cambodia Territory and their tectonic and metallogenic relationships are still controversial (Zaw et al. 2014; Cheng et al. 2019; Kasahara et al. 2021). Hence, there is currently an absence of knowledge regarding the tectonic context and metallogenesis of LFB in Cambodia, representing a gap in our understanding of the geological evolution of the Southeast Asian region. The aim of this research is to study petrology, geochemical characteristics and U–Pb zircon geochronology of the magmatic rocks of the Phnom Sro Ngam Prospect and the Halo Prospect in southwestern and northern Cambodia to understand the petrogenesis, tectonic setting and metallogenic implications for the copper and gold mineralization.

2 | Geological Setting

2.1 | Regional Geological Setting

Cambodia is situated in the southwestern region of the Indochina Peninsula encompassing the mainland of SE Asia. The mainland SE Asia region comprises several Gondwana-derived terranes including Indochina, South China, Sibumasu and West Myanmar Terranes, which assembled and amalgamated by subduction-collision processes during the Late Palaeozoic to Mesozoic (e.g., Metcalfe 2002, 2011, 2013).

Regional tectonic reconstructions indicate that this assembly occurred through a multi-stage process, whereby the South China and Indochina terranes were among the earliest blocks to rift from Gondwana, followed by the rifting and northward drift of the Sibumasu closure and West Myanmar terranes during the Late Palaeozoic. Subsequent subduction and progressive closure of the Paleo-Tethys Ocean beneath the Indochina margin ultimately led to terrane collision and amalgamation during the Triassic Indosinian Orogeny (Metcalfe 2002, 2013; Sone and Metcalfe 2008; Metcalfe 2013). Cambodia is endowed with many mineralized belts including the Truong Son Fold Belt (TSFB), LFB and Dalat-Kratie Belt (DKB), consisting of abundant base metal and precious deposits that are associated with plutonic, volcanic and metamorphic rocks (Figure 1; Barber and Crow 2003; Metcalfe 2013; Burrett et al. 2014; Zaw et al. 2014).

The LFB hosts a variety of mineral deposits, including Cu–Au skarn, Au skarn, epithermal Au and epithermal Sb–Au deposits (Jacobson et al. 1969; Muller 1999; Rodmanee 2000; Kamvong and Zaw 2009; Zaw et al. 2014), Kusa Cu–Au deposit (Kusa, Preah Vihear in Cambodia, Geopacific Resources Ltd 2014). The LFB occupied the region from northern Laos to western Cambodia, through northeastern Thailand and occurred in four tectonic-metallogenic periods such as Ordovician Silurian (450–400 Ma), Late Devonian-Middle Permian (ca. 361–265 Ma), Early-Middle Triassic (250–237 Ma) and Late Triassic (236–200 Ma), with peaks in the Carboniferous and Triassic (Sone and Metcalfe 2008; Kamvong and Zaw 2009; Udchachon et al. 2011; Ueno and Charoentitirat 2011; Khositanont et al. 2013; Zaw et al. 2014; Salam et al. 2014; Qian et al. 2015; Thassanapak et al. 2017; Ouyang et al. 2019; Nie et al. 2019; Kasahara et al. 2021). These tectonic-metallogenic periods broadly correspond to the progressive evolution from early arc development to mature continental arc magmatism associated with Paleo-Tethys subduction and subsequent Indosinian collision, which exert first-order controls on magma generation and mineralization across the LFB (Sone and Metcalfe 2008; Zaw et al. 2014).

Stratigraphically, the LFB consists of volcano-sedimentary successions interbedded with carbonate and siliciclastic units, indicative of deposition along an active continental margin (Zaw et al. 2014; Metcalfe 2013). Typical pre-arc to intra-arc assemblage includes basalt-andesite-dacite volcanic suites and associated pyroclastic deposits and volcaniclastic sedimentary rocks, locally intruded by calc-alkaline diorite-tonalite-granodiorite-granite plutons that present arc a magma plumbing system. Such volcanic-volcaniclastic packages commonly alternate with shallow-marine carbonates and siliciclastic strata in fore-arc basin, supporting the volcanic-arc background interpretation. These sedimentary successions are generally subdivided into a Palaeozoic-Early Mesozoic stratigraphic framework, in which (i) Permian carbonate platforms and carbonate-rich shelf deposits are commonly developed and (ii) Triassic volcanic-sedimentary sequences and associated siliciclastic basinal deposits become increasingly prominent, reflecting intensified arc volcanism and basin development during Paleo-Tethys convergence. Carbonate units are typically thick-bedded to massive limestone/dolostone (locally fossiliferous), whereas the siliciclastic components comprise sandstone-siltstone-shale packages that are commonly interlayered with tuffs and volcaniclastics within the volcano-sedimentary successions, providing a practical basis for stratigraphic division and correlation across the belt (Sone and Metcalfe 2008; Metcalfe 2002; Metcalfe 2013). These features collectively reflect a tectonic setting controlled by subduction-related processes and are temporally associated with the Indosinian orogeny (Sone and Metcalfe 2008; Metcalfe 2002). The geochemical signatures, mineralogical compositions and U–Pb zircon geochronology from these rocks strongly support an origin in a volcanic arc environment, with magma generation involving both fractional crystallization and partial melting of mafic lower crust or amphibolite source rocks (Zaw et al. 2014; Salam et al. 2014). This volcanic-arc interpretation is consistent with Paleo-Tethys reconstruction across mainland SE Asia, including

10991034, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/gj.70352 by Cambodia Hinari NPL, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

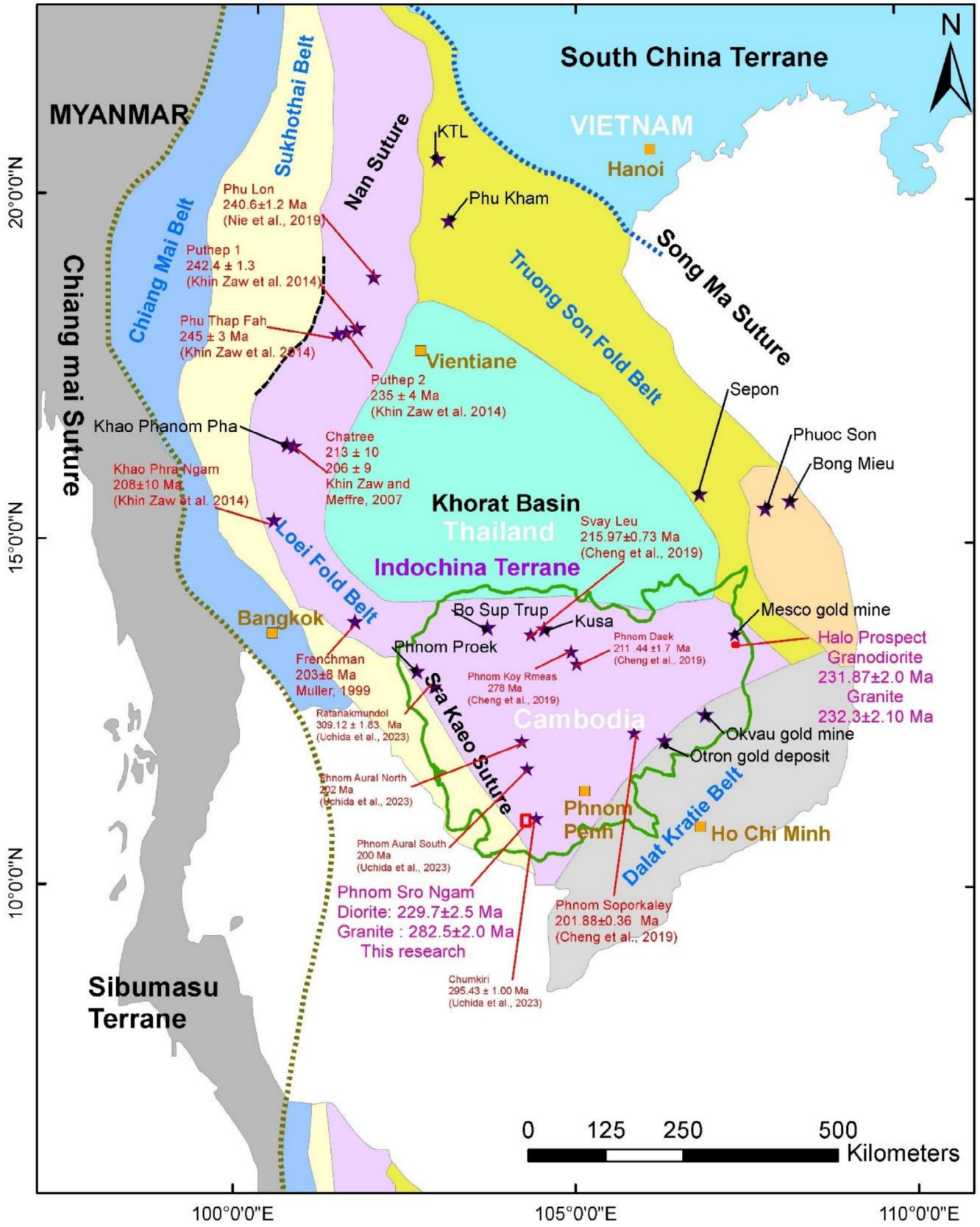


FIGURE 1 | Map showing tectonic divisions of the Indochina Terrane including the mineralized belts of Cambodia, Vietnam, Laos and Thailand (modified after Zaw et al. 2014). SRTM data from http://srtm.csi.cgiar.org/; regional tectonic map modified after for example, Barber and Crow (2003), Metcalfe (2013), Burrett et al. (2014).

10991034, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/gj.70352 by Cambodia Hinari NPL, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

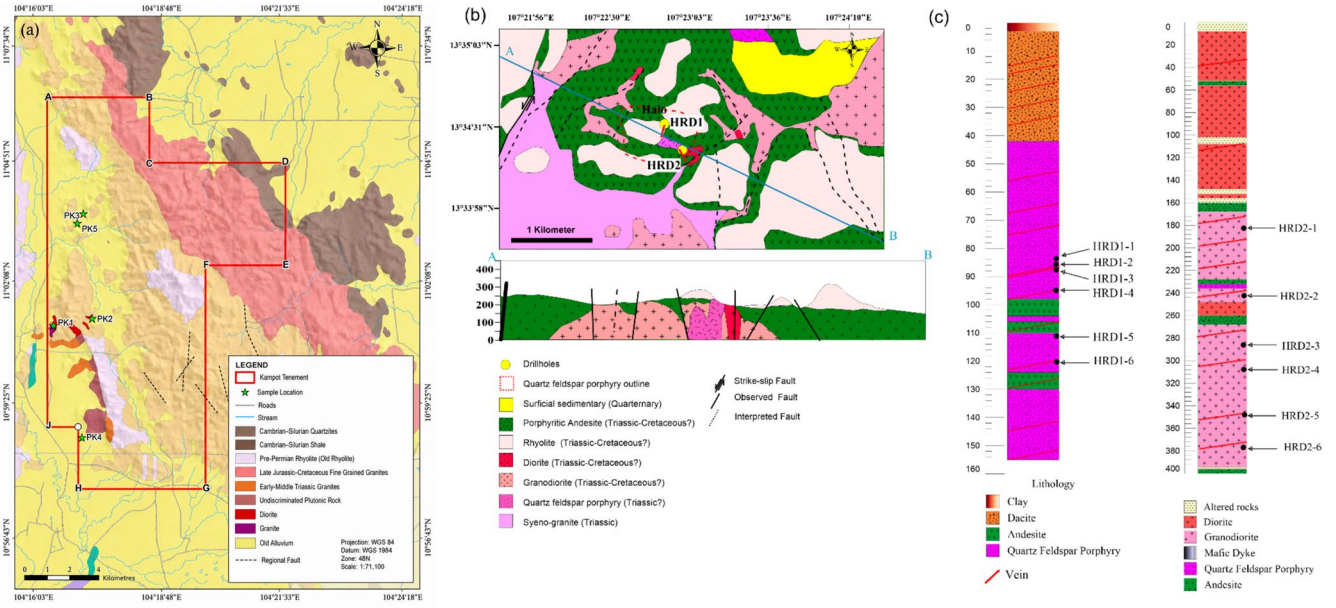


FIGURE 2 | (a) Geological map of the Phnom Sro Ngam Prospect, Southern Cambodia (JICA 2010); (b) Geological map of the Halo Prospect, Northeast Cambodia (Richardson et al. 2016); (c) Core logs of two drill holes at Halo (Seang et al. 2019).

evidence for rifting and subduction processes along the Song Ma tectonic zone in North Laos (Zhang et al. 2021) and Permian arc-related mafic magmatism in NW Thailand (Zhang et al. 2021) and Permian arc related mafic magmatism in NW Thailand (Zhang et al. 2016), as well as Triassic granitoid constraints on the East Paleo-Tethyan assemblage boundary in West Indonesia region (Zhang et al. 2023). These interpretations are consistent with regional tectonic reconstructions and metallogenic studies conducted across Thailand, Laos and Cambodia, which collectively highlight the LFB as an important segment of the Southeast Asian mineral belt (Zaw et al. 2014; Metcalfe 2013; Cheng et al. 2019).

In addition, the geology of northeastern Cambodia is associated with the LFB, characterized by an extensional setting and mantle-derived magmas with adakitic composition. The Late Permian to Triassic plutonic rocks (278–202 Ma) in this region include Phnom Daek, Phnom Koy Rmeas, Oyadav South, Svay Leu and Phnom Soporkaley (Kasahara et al. 2021). This late Permian–Triassic magmatism is temporally consistent with regional arc-related magmatic activity developed during Paleo-Tethys subduction and Indosinian tectonism, reinforcing the interpretation that Cambodia represents the eastern continuation of the LFB magmatic metallogenic system (Metcalfe 2013; Kasahara et al. 2021). Based on geochemical studies, most of the granitoids of LFB are I-type, consistent with the abundance of Cu–Au mineralization in the belt (Zaw et al. 2014).

2.2 | Deposit Geology

The Phnom Sro Ngam Prospect area is covered by quartzite, metasandstone, phyllite, schist, limestone, meta-sedimentary, diorite, granodiorite, granite, rhyolite and dacite porphyry (Figure 2a) (Ly et al. 2022). The gold-bearing veins are hosted by quartzite and schist that have been invaded by magmatically and hydrothermally derived auriferous, sulfide-bearing fluid. The gold mineralization is associated with NW and WNW trending quartz veins and the associated massive sulfide minerals are predominantly pyrite and minor chalcopyrite. Sericitization and silicification halos are evident within the vein structures. Based on the ore microscopy of the polished sections, mineralization consists of pyrite, chalcopyrite, sphalerite, galena, chalcocite, covellite, bornite, azurite and malachite. Vein styles are quartz + pyrite vein and quartz + pyrite + chalcopyrite + galena + sphalerite vein (Samnang Angkor Development Co. Ltd. 2019; Ly et al. 2022).

The Halo Prospect is hosted in felsic and intermediate intrusive and volcanic rocks (Figure 2b,c). The volcanic rocks were intruded by diorite, granodiorite and quartz-feldspar porphyry stock and are associated with the copper and molybdenite mineralization in veins and veinlets. The quartz-feldspar porphyry exposed on the surface is medium- to coarse-grained with abundant disseminated sulfides and appears to be closely related to the molybdenum copper mineralization. A major magnetic, concentrically ringed 'doughnut' anomaly is present in the central part of the Halo porphyry copper-molybdenum Prospect (Richardson et al. 2016). This magnetic anomaly is a typical expression of a porphyry system with a magnetic high in the centre surrounded by a zone of lower magnetic response. The high magnetic anomaly also typically represents an increase in the abundance of magnetite associated with the mineralization in the centre of the porphyry deposit surrounded by low magnetism which reflects a depletion of the abundance of magnetite in the mineralized host rocks (Berger et al. 2008).

Termite mound geochemistry revealed anomalies of copper and molybdenum at the central part of the Halo deposit with a high up to 700 ppm Cu while the anomaly is generally > 250 ppm Cu, covering 1.5 km extending WNW-ESE; 1 km extending NNE–SSW. The high anomaly of molybdenum is 400 ppm in 1.2 km by 75 m, extending EW-NS from the central part of the Halo Prospect. Rock chip geochemistry ranges up to 7470 ppm Cu and up to 1450 ppm Mo at the central part of the Halo area (Angkor Resources Corp 2018; this study). The main alteration zones developed in this

10991034, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/gj.70352 by Cambodia Hinari NPL, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

prospect on the surface are inner phyllic (sericite), outer phyllic (sericite–chlorite), argillic (kaolinite–dickite–pyrophyllite–halloysite), propylitic (chlorite–epidote–quartz–calcite) and silicic quartz alteration according to field observation and short-wave infrared spectroscopy. Phyllic alteration is located in the central part of the Halo Prospect and was overprinted by silicic alteration (outcrop sample on the hilltops, up to 15 m) (Richardson et al. 2016).

3 | Fieldwork, Sampling and Petrography

Quartz-feldspar porphyry samples (HRD1-1, HRD1-2, HRD1-3, HRD1-4, HRD1-5 and HRD1-6) were collected from drill hole HRD1 at depths of 86, 88, 89, 95, 110 and 121 m from drill hole HRD1 in Halo Porphyry Copper-Molybdenum Prospect, Oyadav south, Ratanakiri, with the sampling location shown in Figure 2a (JICA 2010; Richardson et al. 2016; Seang et al. 2019). The granodiorite samples (HRD2-1, HRD2-2, HRD2-3, HRD2-4, HRD2-5 and HRD2-6) were also collected from drill hole HRD2 at depths of 187, 248, 294, 356 and 380 m. Quartz-feldspar porphyry (granite) is yellow to pink in colour and medium- to coarse-grained and consisting of euhedral to subhedral phenocrystic plagioclase (up to 8 mm across recognized in hand specimen), phenocrystic quartz (up to 2.8 mm) and groundmass is mainly of granular quartz (less than 0.3 mm), K-feldspar (less than 0.3 mm across) and biotite (up to 1.4 mm long), with accessory minerals that include zircon and Fe–Ti oxides (main magnetite) (Figure 3a–c). The mineral composition of quartz-feldspar porphyry includes plagioclase (~35%), quartz (~30%), K-feldspar (15%), biotite (~10%), sericite (~5%), chlorite (~3%) and accessory minerals (zircon, Fe–Ti oxides like magnetite ~2%).

The granodiorites are white to dark grey, locally pinkish dark grey and medium- to coarse-grained. They consist mainly of euhedral to subhedral plagioclase phenocrysts (up to 7 mm across), quartz (up to 3.4 mm), K-feldspar (less than 0.3 mm), biotite (less than 1.4 mm across) and hornblende (up to 1.2 mm long) (Figure 3d–i). The mineral composition of granodiorite includes plagioclase (~40%), quartz (~25%), K-feldspar (~10%), biotite (~10%), hornblende (~8%), sericite (~5%) and chlorite (~2%). The quartz-feldspar porphyry and granodiorite show alteration assemblages with feldspar being replaced by sericite, and biotite replaced by chlorite in the groundmass (Figure 3).

Least-altered diorite and granite samples were also collected from outcrops in the Phnom Sro Ngam Prospect, Kampot Province (Figure 2b,c). Granite is light pink in colour and coarse-grained and consists of euhedral to subhedral phenocrystic plagioclase (up to 3 mm across), phenocrystic quartz (up to 15 mm), fine-grained quartz and alkali felspar. Plagioclase was partially altered

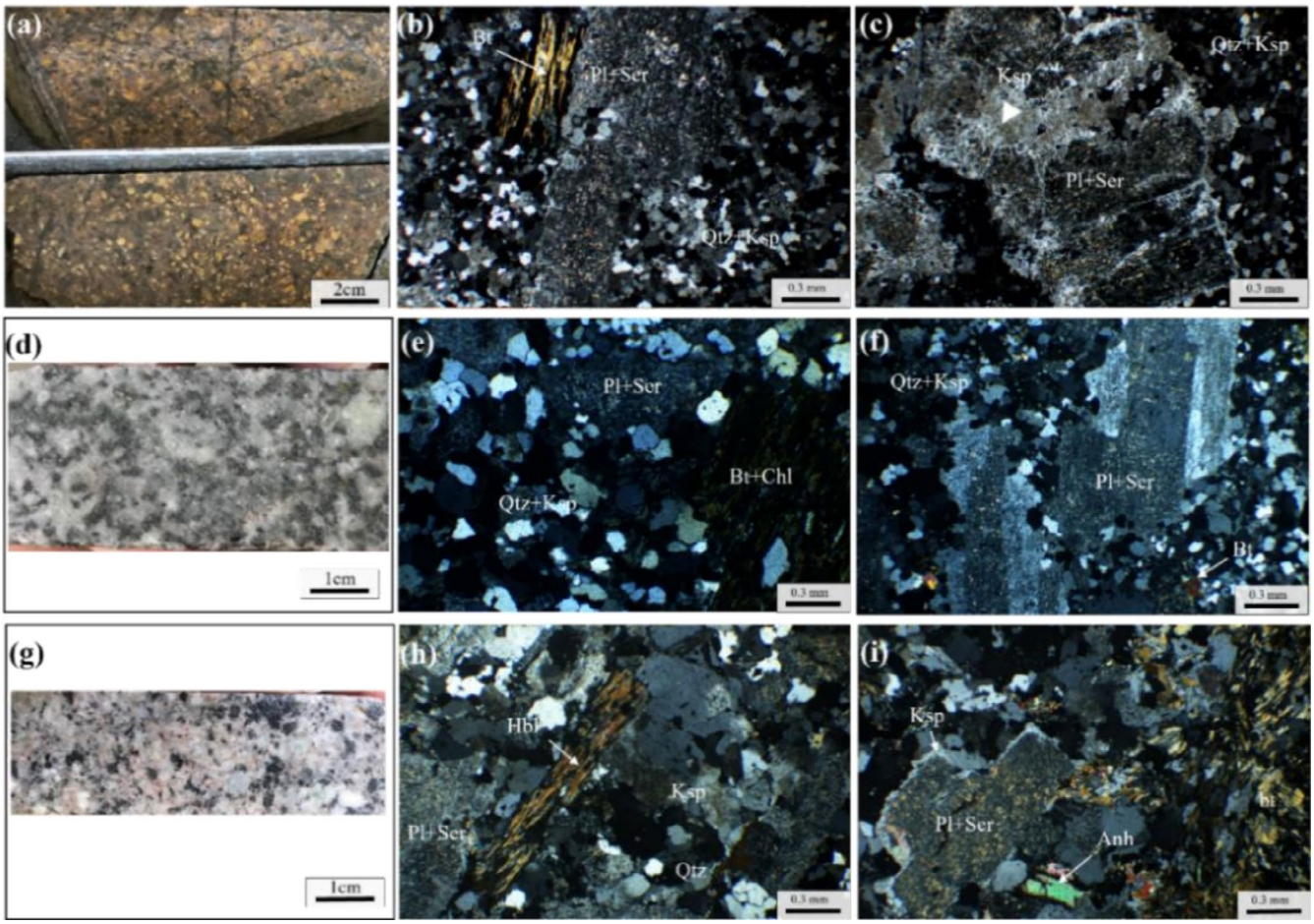


FIGURE 3 | Photographs and photomicrographs show lithological characteristics of plutonic rocks in the Halo Prospect. (a) Hand specimen of granite (quartz-feldspar porphyry). (b) and (c) K-feldspar, plagioclase, quartz, biotite and felsic groundmass in quartz-feldspar porphyry. (d) and (g) Hand specimen of granodiorite, (e), (f), (h) and (i) Plagioclase, hornblende, biotite, quartz and sericite in granodiorite.

10991034, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/gj.70352 by Cambodia Hinari NPL, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

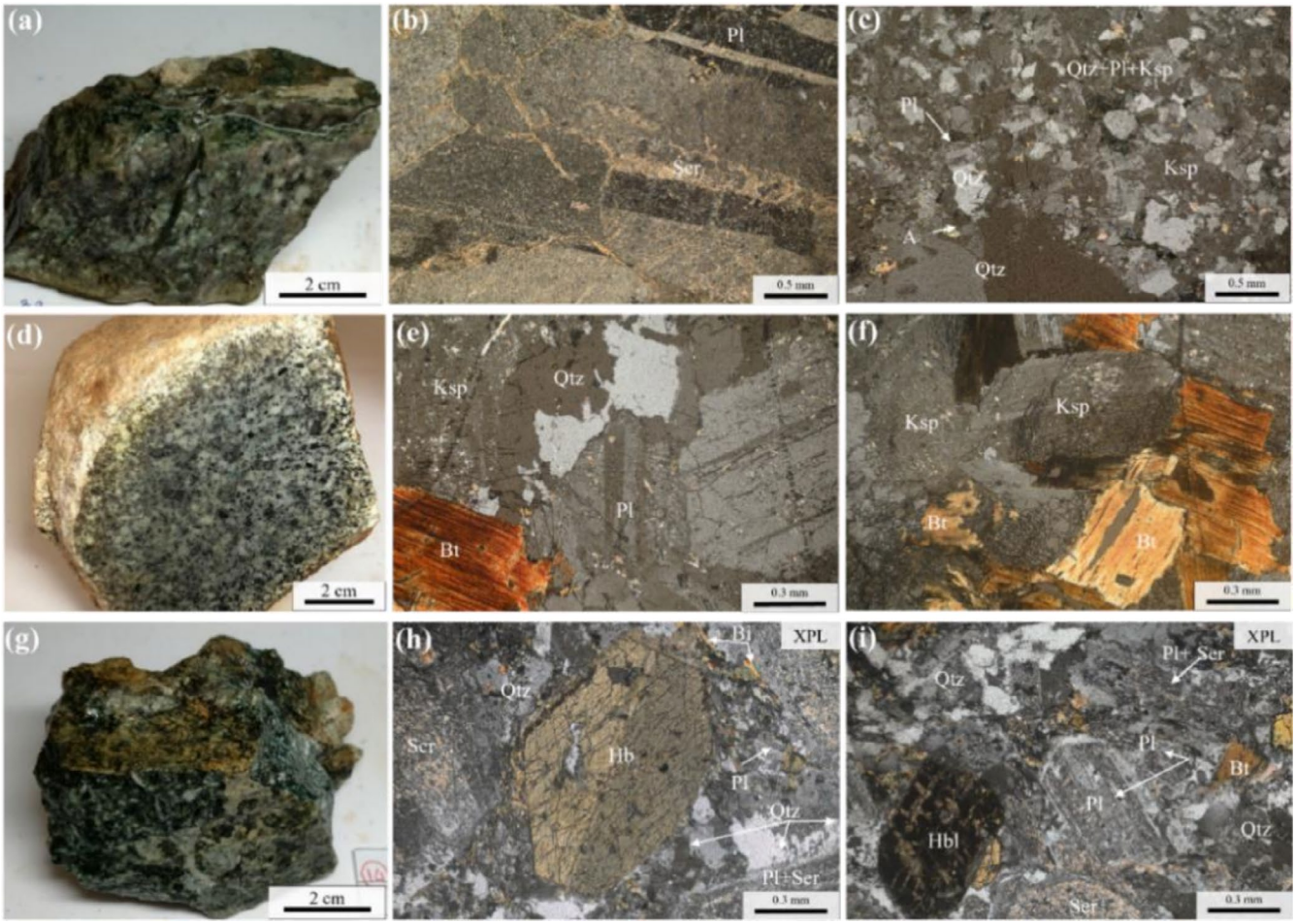


FIGURE 4 | Photographs and photomicrographs showing lithological characteristics of plutonic rocks in the Phnom Sro Ngam Prospect. (a) Hand specimen of granite, (b) and (c) K-feldspar, plagioclase, quartz and felsic groundmass in granite, (d) and (g) hand specimen of diorite, (e), (f), (h) and (i) quartz, K-feldspar, biotite, plagioclase, hornblende and sericite in diorite.

to sericite. The mineral composition of granite includes quartz (~30%), plagioclase (~25%), K-feldspar (~25%), biotite (~15%) and sericite (~5%). Diorite is dark grey to greenish-light grey in colour and medium grained and consists mainly of plagioclase (up to 2 mm across), quartz (less than 1 mm across), K-feldspar (up to 0.6 mm long), euhedral to subhedral hornblende (up to 1.2 mm) and biotite (less than 1 mm across). The mineral composition of diorite includes plagioclase (~50%), quartz (~10%), K-feldspar (~10%), biotite (~15%), hornblende (~10%) and sericite (~5%). Some plagioclases were slightly altered to sericite and biotite was altered to chlorite (Figure 4). Four representative samples such as diorite (PK1), granite (PK5), Quartz-feldspar porphyry (HRD1-8) and granodiorite (HRD2-3) were chosen for zircon U–Pb dating.

3.1 | Laboratory Analytical Methods

3.1.1 | Whole-Rock Geochemistry

Five plutonic rocks were collected from outcrops in Phnom Sro Ngam, Southwestern Cambodia and 12 drill cores of plutonic rocks from the Halo Prospect, Oyadav South, Northeastern Cambodia (Figure 2). Those magmatic rocks were studied using thin petrographic sections to identify textural characteristics and mineral compositions. The ore microscopic features were observed using a Nikon Eclipse E600 at Kyushu University and the Institute of Technology of Cambodia. Whole-rock compositions, including major oxides and trace elements, were conducted by X-ray fluorescence (XRF) spectroscopy using a RIGAKU RIX-3100 at the Department of Earth Resources Engineering and Center of Advanced Instrumental Analysis, Kyushu University. The powder of samples was pressed as pellets at 20 t for about 2 min in a vinyl chloride ring.

Loss on ignition (LOI) was determined by heating the samples to 1000°C for 2 h to get relative weight loss. Precision and accuracy of XRF were based on replicate analysis of the JA-3 standard sample supplied by the Geological Survey of Japan. The measurement condition of XRF analysis was 30 kV and 70 mA. The concentration of rare earth elements (REE) was determined by inductively coupled plasma mass spectrometry (ICP-MS) using Agilent Technologies 7500 at the Center of Advanced Instrument Analysis, Kyushu University, Japan.

3.1.2 | U–Pb Zircon Age Dating Method

LA-ICP-MS analyses of zircon were performed on mounted grain separates. Samples were crushed and non-magnetic heavy minerals were separated at CODES Analytical

10991034, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/gj.70352 by Cambodia Hinari NPL, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Laboratories at the University of Tasmania using gravity and magnetic separation techniques. After heavy mineral and magnetic separation, selected crystals were placed on double-sided sticky tape and epoxy glue was then poured on top of the zircons into a 2.5 cm diameter mould. The epoxy mount was dried for 12 h and polished using clean sandpaper and a clean polishing lap. The samples were then washed in distilled water in an ultrasonic bath. Mounts were kept in a vacuum desiccator overnight to remove moisture and atmospheric air (Thompson et al. 2018).

LA-ICP-MS spot analyses of zircon were performed at the CODES Analytical Laboratories at the University of Tasmania. RESOLution (ASI/Applied Spectra) S-155 ablation system equipped with Coherent Compex Pro 110 ArF excimer laser operating at a 193 nm wavelength and a pulse width of 20 ns. The laser system was coupled to an Agilent 7900 quadrupole ICP mass spectrometer. A laser beam diameter of 20 μm, a repetition rate of 5 Hz and a fluence of 2.1 J/cm^2 were used for analysis. Each spot was ablated with five shots to clean the surface of the mineral of interest. A 30-s measurement of the gas background was performed before a 30-s measurement of each spot ablation.

Ablations were performed in an atmosphere (0.35 L/min flow) and the sample aerosol was mixed with Ar (1.05 L/min flow) straight after ablation. About 2.5 mL/min N2 gas was also added to the sample aerosol to increase instrument sensitivity for heavy isotopes. Isotopes measured were 31P, 49Ti, 56Fe, 89Y, 91Zr, 93Nb, 139La, 140Ce, 141Pr, 146Nd, 147Sm, 153Eu, 157Gd, 159Tb, 163Dy, 165Ho, 166Er, 169Tm, 172Yb, 175Lu, 178Hf, 181Ta, 204Pb, 206Pb, 207Pb, 208Pb, 232Th, 235U and 238U with each element being measured every 0.25 s with longer counting time on the Pb and U isotopes compared to the other elements. ICPMS was tuned to maximize sensitivity while maintaining a U/Th of ≈1.05 during a line scan (3 μm/s) ablation of the NIST612 glass and to minimize oxide production (ThO/Th < 0.15%) after warming up for 2 h.

The data reduction used for U–Pb dating was based on the method outlined in detail in Thompson et al. (2018) and Halpin et al. (2014) using LADR software. For each analysis, a subset of the data most closely matching a concordant composition has been selected for quantification. The downhole fractionation, instrument drift and mass bias correction factors for Pb/U ratios for zircon were calculated using analyses of the 91,500 zircon using values of Wiedenbeck et al. (1995). The instrument drift and mass bias correction factors for the $^{207}Pb/^{206}Pb$ ratio and $^{207}Pb/^{206}Pb$-based ages were calculated using analyses of the NIST610 glass, using the Pb isotopic values of Baker et al. (2004). Trace element abundances in zircon are calibrated on the NIST610 glass using values of Jochum and Enzweiler (2014) and secondary standard corrections based on the compositions of glasses BCR-2G and GSD-1G (GeoReM preferred values; http://georem.mpch mainz.gwdg.de/). Standards were analysed in duplicate at the beginning, end and every 60 min throughout the analytical session. Systematic uncertainties are calculated according to Horstwood et al. (2016).

Quantification was performed using 91Zr as the internal standard element, normalizing all measured cations to the Zr site total of a stoichiometric zircon. The calibration of the U–Pb ages was checked on analyses of the Temora zircon (Black et al. 2003) and the Plešovice zircon (Sláma et al. 2008) analysed throughout the analytical session and treated as unknowns. All common Pb corrections are done using Stacey and Kramers's (1975) model Pb composition at the age of the zircon.

4 | Results

4.1 | Whole-Rock Major and Trace Element Geochemistry

A summary of the major, trace and REE concentrations of plutonic rocks in the Halo and Phnom Sro Ngam Prospects is shown in Table 1. The SiO_2 and K_2O contents ranged between 60.10 and 75.60 wt.% K_2O ranged from 1.14 and 7.16 wt.%, respectively. MnO and P_2O_5 concentrations were less than 0.5 wt.%, and LOI values ranged from 1.16 to 6.18 wt.%. PK1 and HRD2-1 have relatively high LOI values exceeding 5 wt% and slightly lower major-element totals, which may indicate minor post-magmatic alteration. Accordingly, the major-element data for these samples are interpreted in a conservative manner. However, the trace-element data for these samples are generally consistent with those of the other samples, especially in terms of HFSE and REE distributions. As these elements are widely regarded as relatively immobile during low to moderate degree alteration, they are considered to retain the original protolith geochemical signature. Therefore, the tectonic interpretations proposed in this study rely primarily on these relatively immobile trace-element signatures. These data were applied to construct several discrimination and variation diagrams to distinguish the rock types, identify their geochemical characters and constrain the tectonic setting of intrusive magmatism.

Based on whole-rock composition in terms of $Na_2O + K_2O$ and SiO_2 contents, plutonic rocks of the Halo Prospect and Phnom Sro Ngam Prospect are classified as granodiorite and granite and granodiorite and granite and granite (quartz-feldspar porphyry), respectively (Figure 5a) (Cox et al. 1979). The rocks belong to the calc-alkaline series as plotted in the alkali–iron–magnesium (AFM) diagram (Figure 5b) (Irvine and Baragar 1971). In the K_2O versus Si_2O diagram, granodiorite and granite of the Halo Prospect fall into the high-K calc-alkaline to shoshonite. The granite with high-K values contains hydrothermal K-feldspar and biotite, implying that K was added to the rocks during hydrothermal alteration (potassic alteration). The least altered granodiorite belongs to the high-K calc-alkaline series. The diorite and granodiorite of Phnom Sro Ngam fall into the medium calc-alkaline series (Figure 5c). The diorite from Phnom Sro Ngam and granodiorite from the Halo Prospect are metaluminous to peraluminous, with A/CNK (molecular ratio of $Al_2O_3/[Na_2O + K_2O]$) ratios ranging from 0.52 to 1.12. The granite from the Phnom Sro Ngam and Halo Prospects are peraluminous, with A/CNK (molecular ratio of $Al_2O_3/[Na_2O + K_2O]$) ratios ranging from 1.09 to 1.46 (Figure 5d; Maniar and Piccoli 1989).

In addition, the chondrite-normalized REE patterns (Sun and McDonough 1989) of intrusive rocks from Phnom Sro Ngam (diorite and granite) and Halo (granodiorite and granite) show general enrichment of light REEs (LREE) ([(La/Yb)N] = 4.62 to

10991034, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/gj.70352 by Cambodia Hinari NPL, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 | Major, trace and rare earth elements of intrusive rocks from the Phnom Sro Ngam and Halo prospects.

Location	Phnom Sro Ngam prospect, Kampot					Halo prospect, Ratanakiri											
Rock type	Diorite				Granite	Granite (quartz-feldspar porphyry)						Granodiorite					
Sample ID	PK1	PK2	PK3	PK4	PK5	HRD1-1	HRD1-2	HRD1-3	HRD1-4	HRD1-5	HRD1-6	HRD2-1	HRD2-2	HRD2-3	HRD2-4	HRD2-5	HRD2-6
SiO_2 (wt. %)	62.10	61.8	61.95	60.1	72.5	72.41	72.89	74.5	71.36	75.6	73.06	66.85	65.48	65.39	67.45	65.81	66.33
TiO_2	0.47	0.48	0.53	0.45	0.22	0.26	0.3	0.22	0.39	0.27	0.32	0.36	0.38	0.45	0.38	0.41	0.41
Al_2O_3	14.95	15.55	16.35	12.45	13.05	11.99	12.82	11.53	13.49	10.75	11.69	11.3	14.33	14.79	13.8	14.1	14.13
Fe_2O_3T	4.08	5.86	6.07	6.94	2.56	3.46	1.87	2.83	3.02	2.67	3.64	1.84	2.77	3.91	3.34	3.8	3.2
MnO	0.05	0.09	0.11	0.12	0.04	0.02	0.01	0.02	0.04	0.01	0.03	0.02	0.03	0.05	0.04	0.06	0.05
MgO	1.75	2.3	2.92	4.61	0.73	1.03	1.3	0.94	1.61	0.91	1.27	1.56	1.68	2.01	1.45	1.57	1.55
CaO	3.95	5.35	4.38	9.35	1.02	0.09	0.12	0.1	0.21	0.11	0.12	3.63	2.87	2.96	3.2	2.96	3.16
Na_2O	3.18	3.04	3.17	3.55	4.31	0.95	1.15	1.08	1.66	1.11	1.22	1.52	2.1	2.42	2.07	1.99	2.09
K_2O	1.47	1.22	1.79	1.14	2.76	6.02	7.16	6.28	6.47	6.92	6.36	4.81	4.5	3.55	4.34	4.99	4.23
P_2O_5	0.09	0.1	0.1	0.11	0.05	0.02	0.03	0.02	0.04	0.01	0.02	0.07	0.12	0.14	0.13	0.12	0.13
LOI	6.18	4.54	2.67	1.23	1.16	2.59	1.87	1.75	1.65	1.29	1.99	5.59	4.44	3.55	3.04	3.45	3.82
Total	98.27	100.33	100.04	100.05	98.4	98.82	99.53	99.28	99.92	99.65	99.74	97.54	98.68	99.22	99.23	99.25	99.1
Na_2O+K_2O	4.65	4.26	4.96	4.69	7.07	6.96	8.31	7.37	8.13	8.03	7.58	6.32	6.6	5.97	6.41	6.98	6.32
A/CNK	1.07	0.97	1.08	0.52	1.09	1.46	1.3	1.32	1.33	1.13	1.28	0.79	1.06	1.12	0.99	1	1.03
A/NK	2.19	2.46	2.29	1.76	1.29	1.49	1.33	1.34	1.39	1.15	1.31	1.47	1.72	1.89	1.7	1.62	1.76
Ti (ppm)	2793	2852	3150	2674	1307	1533	1777	1331	2306	1575	1919	2127	2228	2662	2270	2413	2442
Ba	250	255	356	470	595	797	929	916	1000	933	859	943	931	882	958	1164	1018
Sr	281	351	279	240	132	90	174	92	222	102	126	337	355	487	460	333	470
Y	13.8	14	16.9	20	17.9	10	6	9	13	6	4	20	18	21	18	19	18
Zr	151	122	124	122	124	124	134	129	156	121	151	133	158	181	169	150	176
Rb	55.2	44.6	70.3	60	86.6	117	123	112	111	111	118	94	97	81	98	108	98
Nb	6.7	6.1	6	5.05	9.6	4	5	4	7	4	5	9	6	7	6	6	6
La	20.7	16.5	17.7	15.6	29.6	11.09	8.97	10.17	11.84	4.29	5.35	71.36	25.83	25.54	33.19	35.13	29.86
Ce	38.8	31.5	32.7	30.4	51.9	20.56	14.95	19.46	23	7.52	9.15	116.25	48.02	47.92	61.65	54.06	54.29

(Continues)

10991034, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/gj.70352 by Cambodia Hinari NPL, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 | (Continued)

Location	Phnom Sro Ngam prospect, Kampot					Halo prospect, Ratanakiri											
Rock type	Diorite				Granite	Granite (quartz-feldspar porphyry)						Granodiorite					
Sample ID	PK1	PK2	PK3	PK4	PK5	HRD1-1	HRD1-2	HRD1-3	HRD1-4	HRD1-5	HRD1-6	HRD2-1	HRD2-2	HRD2-3	HRD2-4	HRD2-5	HRD2-6
Pr	4.69	3.84	4.09	3.79	5.78	2.34	1.67	2.31	2.8	0.76	0.91	12.9	5.15	4.48	6.01	5.47	5.27
Nd	16.9	14.1	14.6	15.8	18.5	7.61	4.78	7.41	9.36	2.75	3.25	35.95	15.92	17.45	21.4	18.62	19.37
Sm	3.42	2.9	2.99	4.01	3.18	1.48	0.93	1.36	1.85	0.49	0.55	5.46	2.76	3.13	3.71	3	3.38
Eu	1.06	0.85	0.9	0.96	0.71	0.57	0.47	0.53	0.58	0.36	0.39	1.24	0.82	0.71	0.92	0.75	1.02
Gd	2.84	2.71	3.01	4.39	2.89	1.5	0.76	1.22	1.8	0.51	0.6	5.66	2.51	2.57	3.31	2.68	3.5
Tb	0.43	0.42	0.47	0.74	0.46	0.22	0.12	0.17	0.26	0.06	0.07	0.62	0.34	0.38	0.46	0.34	0.48
Dy	2.5	2.43	2.92	4.62	2.84	1.15	0.55	0.9	1.42	0.36	0.35	2.93	1.81	2.12	2.39	1.74	2.32
Ho	0.51	0.52	0.62	0.96	0.6	0.25	0.12	0.17	0.25	0.11	0.07	0.55	0.38	0.42	0.47	0.34	0.49
Er	1.46	1.56	1.77	2.75	1.87	0.71	0.32	0.49	0.68	0.29	0.25	1.77	1.1	1.31	1.49	1.12	1.51
Tm	0.21	0.22	0.27	0.4	0.29	0.08	0.03	0.06	0.09	0.04	0.04	0.22	0.15	0.18	0.22	0.15	0.22
Yb	1.38	1.5	1.73	2.42	2.05	0.63	0.36	0.46	0.62	0.29	0.24	1.3	1.09	1.24	1.38	0.98	1.37
Lu	0.21	0.23	0.28	0.4	0.33	0.09	0.05	0.07	0.09	0.05	0.05	0.21	0.18	0.2	0.23	0.17	0.23
Total REE	95.11	79.28	84.05	87.24	121	48.29	34.07	44.8	54.65	17.89	21.26	256.42	106.05	107.65	136.81	124.54	123.33
(La/Yb)N	10.76	7.89	7.34	4.62	10.36	12.56	17.84	15.92	13.7	10.75	16.24	39.3	16.96	14.79	17.24	25.81	15.58
(La/Sm)N	3.91	3.67	3.82	2.51	6.01	4.84	6.25	4.82	4.13	5.66	6.26	8.44	6.04	5.27	5.78	7.57	5.7
(Gd/Yb)N	1.7	1.49	1.44	1.5	1.17	1.96	1.75	2.21	2.4	1.48	2.1	3.59	1.9	1.72	1.98	2.27	2.11
Eu/Eu*	1.01	0.91	0.91	0.7	0.7	1.15	1.67	1.23	0.96	2.19	2.07	0.67	0.93	0.74	0.78	0.79	0.9

10991034, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/gj.70352 by Cambodia Hinari NPL, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

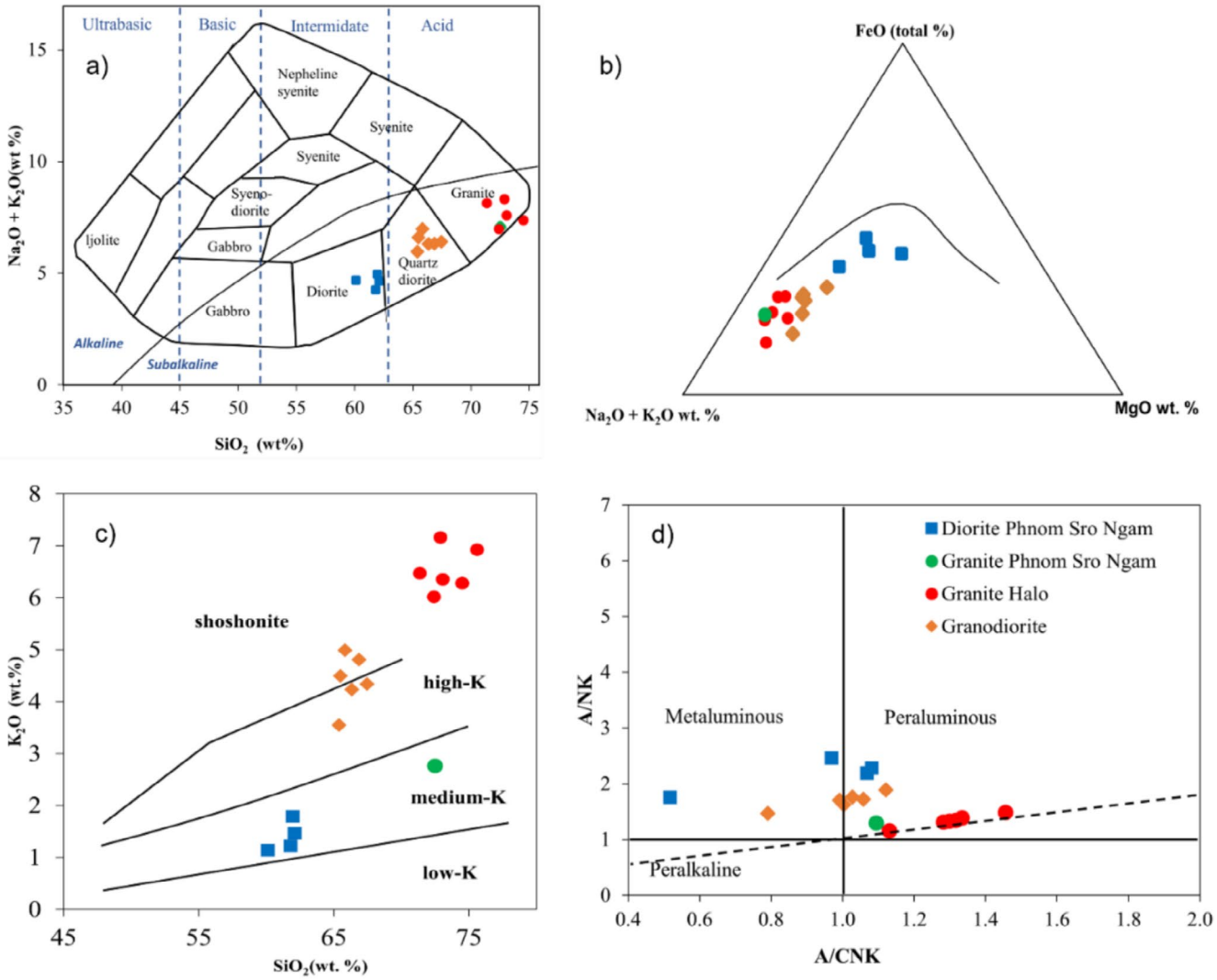


FIGURE 5 | Major elements diagram for the Phnom Sro Ngam and Halo Prospects. (a) TAS diagram for plutonic rocks using SiO_2 versus ($Na_2O + K_2O$), after Cox et al. (1979); (b) AFM diagram showing whole-rock composition in terms of $Na_2O + K_2O$, total iron as FeO and MgO (Irvine and Baragar 1971), (c) SiO_2 versus K_2O diagram, (d) A/CNK versus A/NK plot (A = Al_2O_3, C = CaO, N = Na_2O and K = K_2O in a molar, after Maniar and Piccoli 1989).

10.76) and REEs (LREE) ([(La/Yb)N] = 4.62 to 10.76), respectively and depletion of heavy REEs (HREE) ([(Gd/Yb)N] = 1.17 to 1.70) and REEs (HREE) ([(Gd/Yb)N] = 1.48 to 3.59), respectively (Figure 6a,c). The distinctive weakly negative Eu anomaly (Eu/*Eu = 0.70 to 1.01) of Phnom Sro Ngam diorite and granite and (Eu/*Eu = 0.67 to 0.93) of the Halo granodiorite would be indicative of plagioclase fractionation.

The Halo granite (quartz-feldspar porphyry) is characterized by positive Eu anomalies (Eu/Eu* = 0.96 to 2.19). The positive Eu anomalies suggest that Eu present in primary plagioclase is immobile during potassic alteration, while other REEs were slightly removed during potassic alteration. Thus, REE patterns of the Halo granite were slightly mobile during alteration. The primitive mantle-normalized trace elements diagram of the magmatic rocks in Phnom Sro Ngam and Halo is also characterized by enrichments of large-ion lithophile elements (LILE, such as K, Rb and Ba) depletions of high-field-strength elements (HFSE, such as Nb and Ti) (Figure 6).

4.2 | Zircon U–Pb Age

U–Pb age data obtained from zircon from the diorite and granite from Phnom Sro Ngam and the granodiorite and quartz-feldspar porphyry from the Halo Prospect are presented in Table S1.

The CL images of the zircon grains from samples PK1, PK5, HRD2-3 and HRD1-6 are shown in Figure 7. They are mostly 30 to 100 m in size and exhibit clear oscillatory zones indicating magmatic zircons. Most grains show a clear core-rim texture and well-developed zoning. For the intrusive rocks of Phnom Sro Ngam, 20 grains from diorite were analysed for sample PK1. All of them are concordant and the group of 12 youngest analyses forms a coherent population with the $^{206}Pb/^{238}U$ weighted mean age of 229.7 ± 2.5 (n = 12, MSWD = 1.5, probability of fit 0.11). For granite from Phnom Sro Ngam (PK5), 20 grains were analysed and all of them are concordant (Figure 8a,b).

10991034, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/gj.70352 by Cambodia Hinari NPL, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

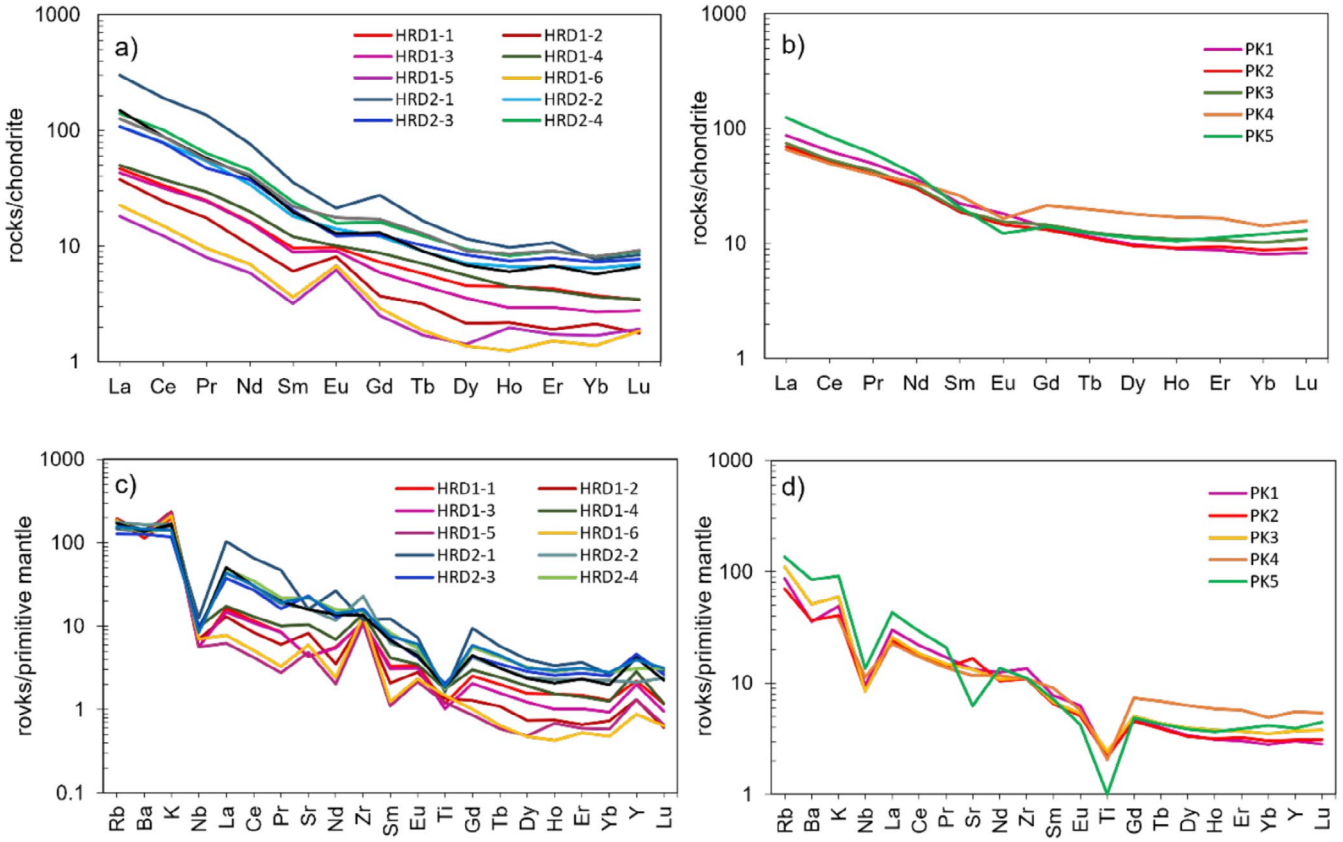


FIGURE 6 | (a) and (b) Chondrite-normalized rare earth element patterns; (c) and (d) primitive mantle (PM)- normalized trace element diagrams for the Phnom Sro Ngam and Halo Prospects (Sun and McDonough 1989).

A group of 12 youngest analyses forms a coherent population with the $^{206}Pb/^{238}U$ weighted mean age of 282.5 ± 2.0 (n = 12, MSWD = 1.12, probability of fit 0.34) and another 8 grains forming an older population with the $^{206}Pb/^{238}U$ weighted mean age of 294.3 ± 2.5 Ma. Intrusive rocks from Halo Prospect: 20 grains from granite porphyry (HRD1-6) were analysed and all of them are concordant (Figure 8). Three analyses contained domains with different ages. A group of 15 youngest analyses forms a coherent population with the $^{206}Pb/^{238}U$ weighted mean age of 232.3 ± 2.10 (n = 15, MSWD = 1.3, probability of fit 0.20) and another seven analyses forming an older population with the $^{206}Pb/^{238}U$ weighted mean age of 241.6 ± 2.5 Ma. Twenty zircon grains from granodiorite (HRD2-3) were analysed and nineteen of them have concordant compositions (Figure 8b,c). A group of 17 youngest concordant analyses forms a coherent population with the $^{206}Pb/^{238}U$ weighted mean age of 231.8 ± 2.0 (n = 17, MSWD = 1.4, probability of fit 0.14). The zircon U–Pb data of intrusive rocks from the Halo and Phnom Sro Ngam Prospects indicate a Triassic to Late Permian crystallization age.

5 | Discussion

5.1 | Petrogenesis and Magma Sources

The intrusive rocks are classified as I-type, M-type and S-type granites in the (Zr + Nb + Ce + Y) versus (Fe_2O_3/MgO) diagram (Figure 9a). Additionally, based on the ($Na_2O + K_2O$) versus (Zr + Nb + Ce + Y) diagram (Figure 9b), the rocks fall within the I- and S-type granite fields (Pearce et al. 1984). Specifically, the diorite from the Phnom Sro Ngam Prospect and the granodiorite Halo Prospect are metaluminous to peraluminous and medium calc-alkaline and high calc-alkaline, respectively, whereas the granite of the Phnom Sro Ngam and Halo Prospects are peraluminous with medium calc-alkaline and shoshonite, respectively. These geochemical characteristics, together with the enrichment of LILE and LREE and the depletion of Nb and Ti, indicate derivation from subduction-modified mantle sources with variable crustal input during magma evolution. According to petrographic observations, the intrusive rocks are composed predominantly of quartz, plagioclase, K-feldspar, biotite and hornblende, which is consistent with hydrous calc-alkaline magmas generated in an arc-related setting. The more felsic and locally peraluminous compositions of the granitic rocks may additionally reflect the involvement of crustal materials during magmatic differentiation and emplacement.

The Linear correlations between MgO, Fe_2O_3, Al_2O_3, TiO_2, CaO, Na_2O and SiO_2 for Phnom Sro Ngam and Halo prospect rocks indicate that fractionation occurred (Figure 10). This interpretation is also supported by petrographic observations, which show hornblende and biotite in the less evolved rocks and more abundant quartz and K-feldspar in the more evolved felsic rocks. The decreasing MgO, Fe_2O_3T and Al_2O_3 with SiO_2 suggests the fractionation of mafic minerals (e.g., amphibole), whilst the decreasing of TiO_2 with SiO_2 indicates the fractionation of Ti oxides (e.g., rutile and ilmenite), as also supported by the distinct negative anomalies of Ti and Nb. Moreover, the negative correlation

10991034, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/gj.70352 by Cambodia Hinari NPL, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

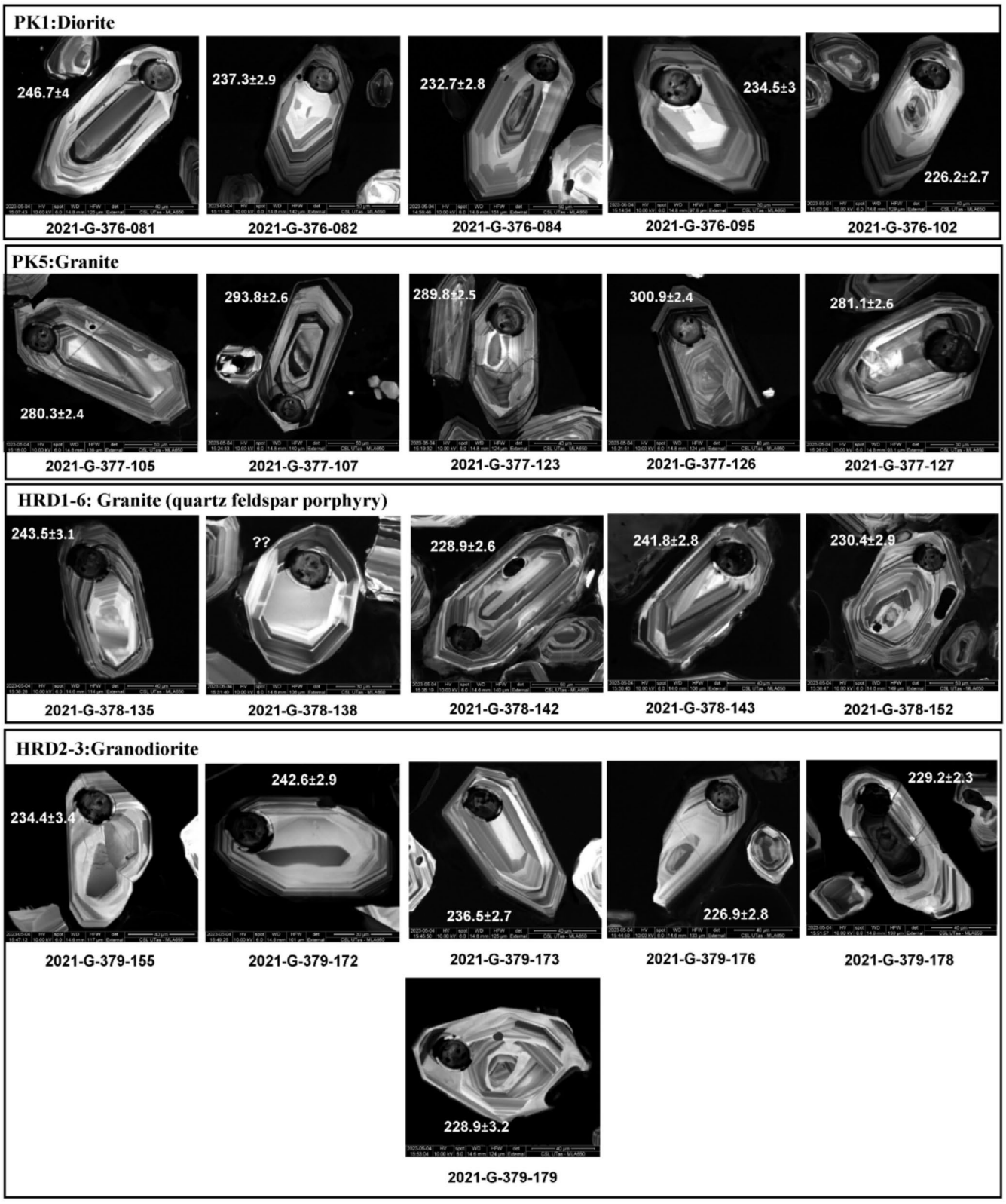


FIGURE 7 | Cathodoluminescence (CL) image of representative zircon from intrusive rocks in Phnom Sro Ngam Prospect (PK1 and PK2) and Halo Prospect (HRD2-3 and HRD1-6).

between CaO and Na_2O with SiO_2 and the distinct negative Eu, Sr and Ba anomalies suggest the significant fractionation of plagioclase. Based on geochemical trends and mineral assemblages indicate that the compositional variation from diorite/granodiorite to granite was controlled by fractional crystallization, with variable crustal contribution.

10991034, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/gj.70352 by Cambodia Hinari NPL, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

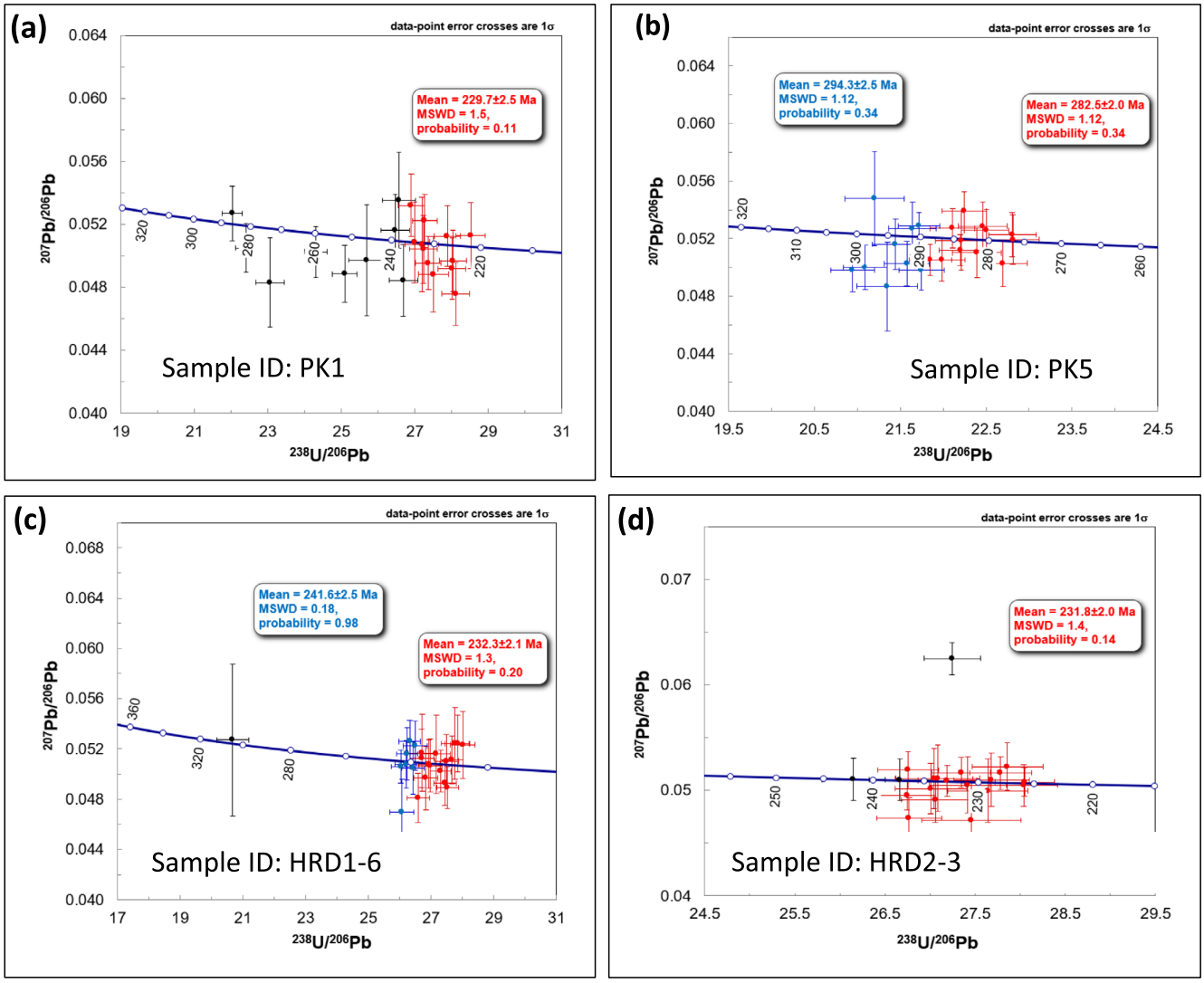


FIGURE 8 | Zircon U–Pb age dating results for the (a) Diorite (PK1) in Phnom Sro Ngam Prospect, (b) Granite (PK5) in Phnom Sro Ngam Prospect, (c) Granite porphyry (HRD1-6) in Halo porphyry and (d) Granodiorite (HRD2-3) in Halo Prospect.

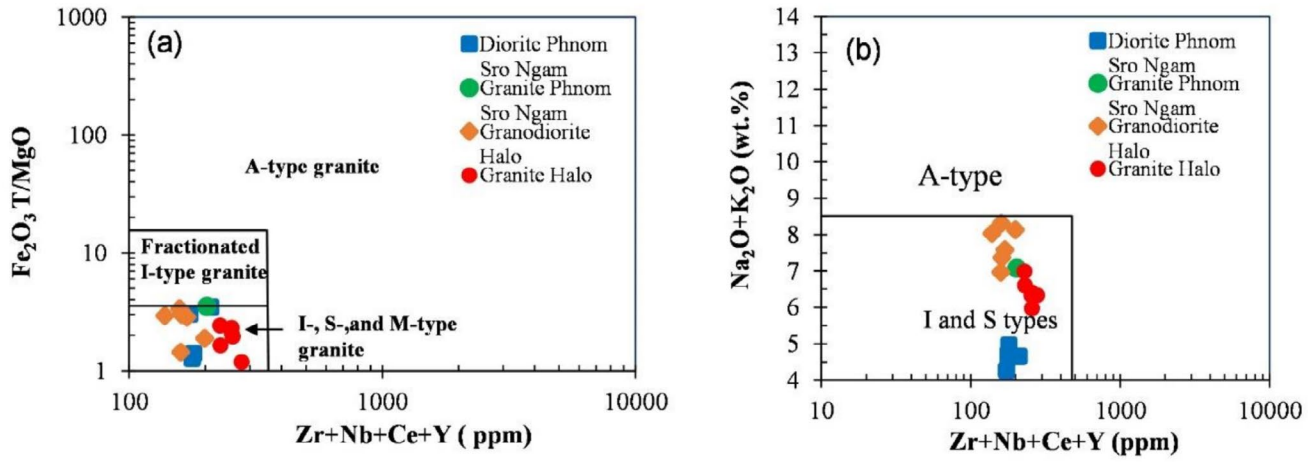


FIGURE 9 | (a) (Zr + Nb + Ce + Y) versus (Fe_2O_3/MgO) diagram (Whalen et al. 1987), ($Na_2O + K_2O$) versus (Zr + Nb + Ce + Y) diagram (Pearce et al. 1984).

10991034, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/gj.70352 by Cambodia Hinari NPL, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

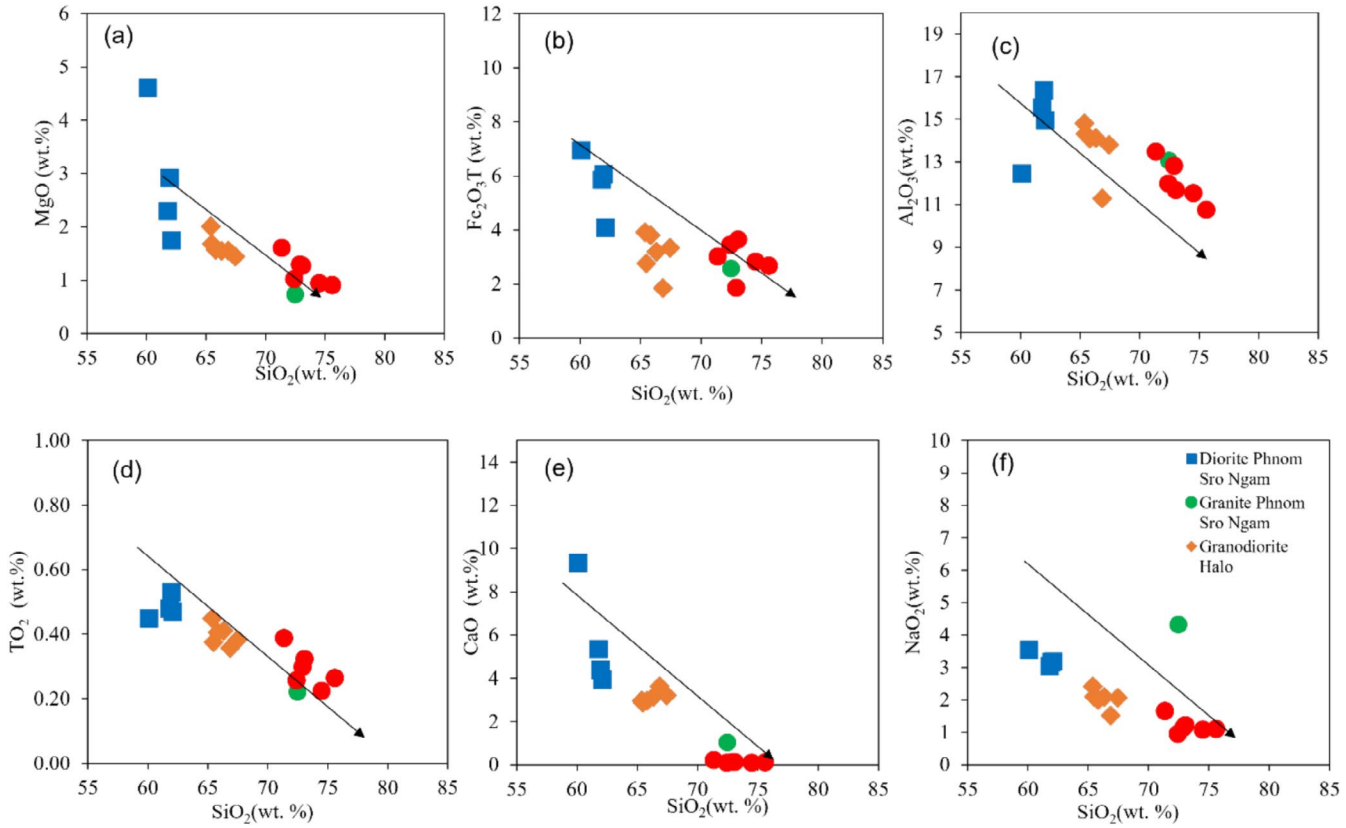


FIGURE 10 | Harker diagrams for the Halo and Phnom Sro Ngam intermediate-felsic magmatic rocks. (a) SiO_2 versus MgO; (b) SiO_2 versus Fe_2O_3T; (c) SiO_2 versus Al_2O_3; (d) SiO_2 versus TiO_2; (e) SiO_2 versus CaO; (f) SiO_2 versus Na_2O.

TABLE 2 | Geochronological data rocks from Phnom Sro Ngam and Halo prospects together with published data from Loei Fold Belt.

Location	Rock type	Age (Ma)	Method	References
Phnom Sro Ngam, Kampot	Diorite	229.7 ± 2.5	U–Pb LA-ICP-MS	This Study
	Granite	282.5 ± 2.0	U–Pb LA-ICP-MS	This Study
Halo, Ratanakiri	Granodiorite	231.87 ± 2.0	U–Pb LA-ICP-MS	This Study
	Granite (quartz-feldspar porphyry)	232.3 ± 2.10	U–Pb LA-ICP-MS	This Study
Svay Leu	Granodiorite	215.97 ± 0.73	U–Pb LA-ICP-MS	Cheng et al. (2019)
Phnom Soporkaley	Granodiorite	201.88 ± 0.36	U–Pb LA-ICP-MS	Cheng et al. (2019)
Phnom Daek	Granodiorite	211.44 ± 1.7	U–Pb LA-ICP-MS	Cheng et al. (2019)
Chum Kiri	Granite	295.43 ± 1.00	U–Pb LA-ICP-MS	Uchida et al. (2023)
Phnom Aural North	Diorite	202	U–Pb LA-ICP-MS	Uchida et al. (2023)
Phnom Aural South	Granite	200	U–Pb LA-ICP-MS	Uchida et al. (2023)
Ratanak Mundol	Granite	309.12 ± 1.83	U–Pb LA-ICP-MS	Uchida et al. (2023)
Porphyry-related skarn Phu Lon Cu–Au deposit, Loei Fold Belt	Diorite	240.6 ± 1.2	U–Pb LA-ICP-MS	Nie et al. (2019)
Pu Thep Cu prospect (PUT2)	Diorite	242.4 ± 1.3	U–Pb LA-ICP-MS	Zaw et al. (2014)
Phu Thap Fah	Granodiorite	245 ± 3	U–Pb LA-ICP-MS	Zaw et al. (2014)
Chatree Au-Ag deposit	Diorite porphyry	213 ± 10 206 ± 9	U–Pb LA-ICP-MS	
Ban Bothong–Khao Phra Ngam	Granodiorite	208 ± 10	U–Pb LA-ICP-MS	Zaw et al. (2014)

Note: For this study, the age data reported in this table represent younger zircon age populations.

10991034, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/gj.70352 by Cambodia Hinari NPL, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

5.2 | Timing of Magmatism

The U–Pb ages of zircons from two intrusive rocks from the Halo Prospect in Ratanakiri, Northeastern Cambodia and two samples from Phnom Sro Ngam in Kampot, Southern Cambodia together with geochronology data from Thailand are shown in Table 2. Geochronological data are important to contain the magmatic event and further understand the tectonic evolution of the Halo prospect and Phnom Sro Ngam prospect. In this study, the zircon U–Pb ages from diorite and granite of Phnom Sro Ngam Prospects are 229.7 ± 2.5 and 282.5 ± 2.0 Ma, respectively. There is also an older population dated 294.3 Ma.

The Zircon U–Pb ages from granodiorite and granite (quartz-feldspar porphyry) from the Halo prospect in Ratanakiri are 231.8 ± 2.0 and 232.3 ± 2.1 Ma, respectively (Figure 8). Additionally, there is an older population with an age of 241.6 Ma.

These ages of the rock in the Halo prospect in Ratanakiri indicate a clear temporal match with the Triassic metallogenic framework, with the main intrusive events at approximately 232–229 Ma falling within the Late Triassic (236–200 Ma) mineralization window recognized in the LFB. The older age of 241.6 Ma probably indicates another magmatic event in the Middle Triassic period (Ladinian).

In addition, the age of Phnom Sro Ngam and Halo prospect is Triassic to late Permian which ranges from 229 to 282 Ma. This timeframe aligns with significant tectonomagmatic events in the region, particularly the Indosinian Orogeny, which resulted from the collision and amalgamation of the Sibumasu and Indochina blocks during the closure of the Paleo-Tethys Ocean (Metcalfe 2002; Sone and Metcalfe 2008; Uchida et al. 2023).

5.3 | Tectonic, Metallogenic and Exploration Implications

All magmatic rocks plot largely in volcanic arc granite field in the (Y + Nb) versus Rb and Y versus Nb tectonic discrimination diagrams (Figure 11a,b) (Pearce et al. 1984). Furthermore, Rb/Zr versus Nb diagram, the intrusive rocks plot within the island arc setting (Figure 11c) (Brown et al. 1984). Based on the $(La/Yb)_N$ versus Yb_N diagram, the diorite and granodiorite plot in the field of post-Archean subduction-related granitoids (Figure 11d; Martin 1986).

The zircon geochemistry has been used by many researchers to decipher barren from fertile magmas to explore for porphyry copper deposits (Lu et al. 2016; Loader et al. 2017; Zou et al. 2019; Loucks et al. 2020; Leslie et al. 2022). Several different thresholds have been used to discriminate intrusions that are fertile and non-fertile for porphyry deposits (e.g., Eu/Eu* > 0.4 Ballard et al. 2002; Dilles et al. 2015; Pizarro et al. 2020; Nevolko et al. 2021; Eu/Eu* > 0.3, 10,000*(Eu/Eu*)/Y > 1, (Ce/Nd)/Y > 0.01 and lower Dy/Yb < 0.3; Lu et al. 2016). The results indicate that samples HRD1-6 and HRD2-3 have ratios of Eu/Eu* > 0.4, Dy/Yb < 0.3 and (Ce/Nd)/Y > 0.01 that

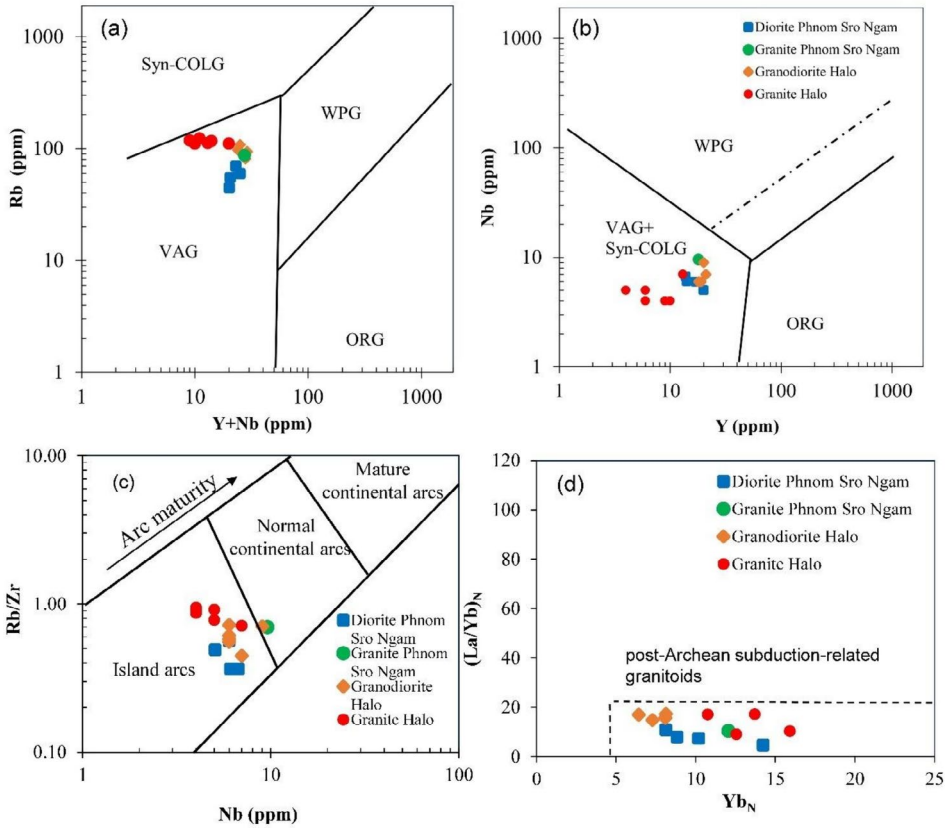


FIGURE 11 | Tectonic discrimination diagrams for plutonic rock in the Halo and Phnom Sro Ngam Prospects: (a) Y + Nb versus Rb and (b) Y versus Nb (after Pearce et al. 1984), (c) Rb/Zr versus Nb diagram (Brown et al. 1984), (d) La/Yb versus Yb with fields from Martin (1986). ORG, ocean-ridge granite; syn-COL G, syn-collision granite; VAG, volcanic arc granite; WPG, within-plate granite.

10991034, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/gj.70352 by Cambodia Hinari NPL, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

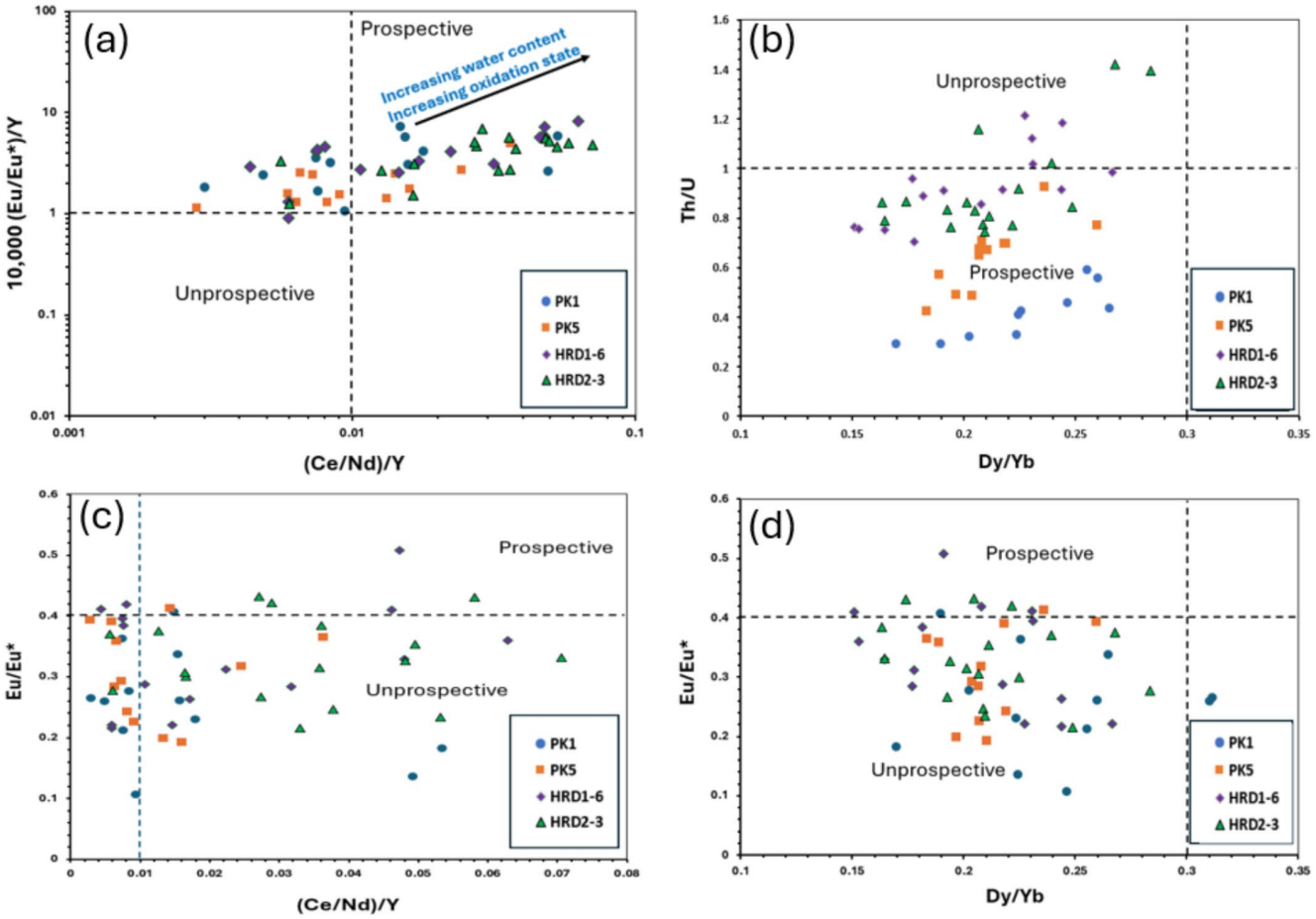


FIGURE 12 | Zircon trace element geochemistry plots for porphyry copper fertility. (a) 10,000 (Eu/Eu*)/Y versus (Ce/Nd)/Y; (b) Th/U versus Dy/Yb; (c) Eu/Eu* versus (Ce/Nd)/Y; (d) Eu/Eu* versus Dy/Yb.

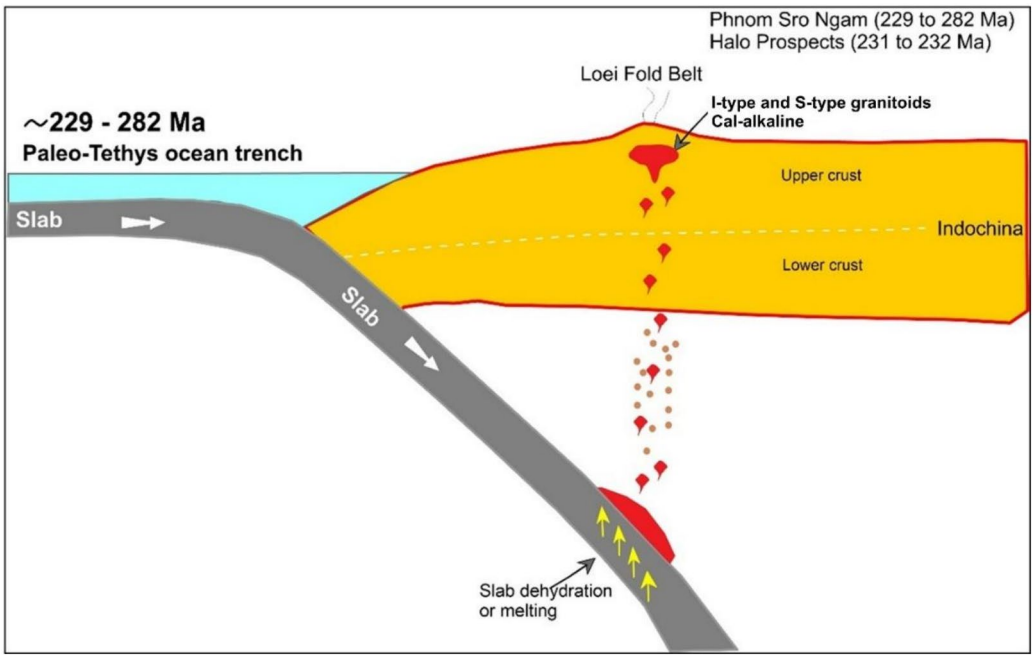


FIGURE 13 | Schematic tectonic evolution of Halo and Phnom Sro Ngam Prospects in Loei Fold Belt during Late Permian to Triassic (modified after Zaw et al. 2014; Kasahara et al. 2021).

10991034, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/gj.70352 by Cambodia Hinari NPL, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

show characteristics of fertile magma for porphyry copper exploration (Figure 12). The link between the igneous rock samples and regional mineralization is not conclusive and should be viewed only as a preliminary indicator or potential association. The metallogenic origins of the Phnom Sro Ngam and Halo Prospects can be traced back to a period spanning from the Triassic to the Late Permian, covering a timeframe of approximately 229 to 282 million years ago. The host magmatic units are associated with volcanic activity in the LFB, which occurred during the subduction of the Paleo-Tethys Oceanic crust towards the east beneath the Indochina terrace (Figure 13; Zaw et al. 2014). The LFB hosts various mineral deposits such as porphyry, skarn and epithermal deposits and these deposits host metals such as gold, copper and other associated minerals (Table 2). Given the correlation between Cambodian igneous rock ages and productive porphyry systems in Thailand, Cambodia represents a high-potential region for future mining exploration.

6 | Conclusions

1. The intrusive and volcanic rocks belong to medium-K volcanic arcs, and I-type and S-type granite. There is an enrichment in large-ion lithophile elements such as Rb, Ba and K and depletion in some high-field-strength elements such as Nb and Ti, suggesting the magma was generated in a subduction-related tectonic setting.
2. LA-ICP-MS zircon U–Pb dating yielded concordant ages ranging from 229.7 ± 2.5 Ma to 282.5 ± 2.0 Ma for the intrusive rocks from the Phnom Sro Ngam Prospect and 231.8 ± 2.0 Ma to 232.3 ± 2.1 Ma for the Halo Prospect intrusive rocks. The intrusive rocks were emplaced during the Triassic to Late Permian, and they correspond to magmatic activity in the LFB.
3. Preliminary fertility study indicates that samples D1-QFP and D2-GR, which have ratios of $Eu/Eu^* > 0.4$, $Dy/Yb < 0.3$ and $(Ce/Nd)/Y > 0.01$, point to fertile magmas for porphyry copper mineralisation.

Acknowledgements

We gratefully acknowledge the support of the Laboratory-Based Education (LBE) promoted by the Project for Strengthening Engineering Education and Research for Industrial Development in Cambodia, Institute of Technology of Cambodia (ITC) and Department of Geology, Ministry of Mines and Energy, Cambodia. We would like to express our deepest thanks to His Excellency Mr. Kong Samnang, Chairman of Samnang Angkor Development Co. Ltd., for allowing and supporting the field research in Phnom Sro Ngam, Chhouk District, Kampot, Cambodia. Moreover, we would like to express our sincere gratitude to Mr. Mike Weeks, Chairman of Angkor Gold, for his permission and support in conducting field research at the Halo Prospect, Oyadav South area, Ratanakiri Province, Cambodia. Open access publishing facilitated by University of Tasmania, as part of the Wiley - University of Tasmania agreement via the Council of Australasian University Librarians.

Funding

The authors have nothing to report.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Peer Review

For transparency, the peer review documents associated with this article are available at https://doi.org/10.1002/gj.70352.

References

Angkor Resources Corp. 2018. "Report of the-Geological Overview of Angkor Gold Projects, Cambodia." 6–35. http://www.angkorgold.ca/wp-content/uploads/2014/06/GEO-PPTMarch-2018.pd.

Baker, J., D. Peate, T. Waight, and C. Meyzen. 2004. "Pb Isotopic Analysis of Standards and Samples Using a 207Pb–204Pb Double Spike and Thallium to Correct for Mass Bias With a Double-Focusing MC-ICP-MS." *Chemical Geology* 211, no. 3–4: 275–303.

Ballard, J. R., J. M. Palin, and I. H. Campbell. 2002. "Relative Oxidation States of Magmas Inferred From Ce(IV)/Ce(III) in Zircon: Application to Porphyry Copper Deposits of Northern Chile." *Contributions to Mineralogy and Petrology* 144, no. 34: 7–364. https://doi.org/10.1007/s00410-002-0402-5.

Barber, A. J., and M. J. Crow. 2003. "An Evaluation of Plate Tectonic Models for the Development of Sumatra." *Gondwana Research* 6: 1–28.

Berger, B. R., R. A. Ayuso, J. C. Wynn, and R. R. Seal. 2008. "Preliminary Model of Porphyry Copper Deposits." In *US Geological Survey Open-File Report Geological Survey*. US Geological Survey. https://doi.org/10.3133/ofr20081321.

Black, L. P., S. L. Kamo, C. M. Allen, et al. 2003. "TEMORA 1: A New Zircon Standard for Phanerozoic U–Pb Geochronology." *Chemical Geology* 200, no. 1–2: 155–170.

Brown, G. C., R. S. Thorpe, and P. C. Webb. 1984. "The Geochemical Characteristics of Granitoids in Contrasting Arcs and Comments on Magma Sources." *Journal of the Geological Society* 141, no. 3: 413–426.

Burrett, C., K. Zaw, S. Meffre, et al. 2014. "The Configuration of Greater Gondwana—Evidence From LA ICPMS, U–Pb Geochronology of Detrital Zircons From the Palaeozoic and Mesozoic of Southeast Asia and China." *Gondwana Research* 26, no. 1: 31–51.

Cheng, R., E. Uchida, M. Katayose, et al. 2019. "Petrogenesis and Tectonic Setting of Late Paleozoic to Late Mesozoic Igneous Rocks in Cambodia." *Journal of Asian Earth Sciences* 185: 104046.

Cox, K. G., J. D. Bell, and R. J. Pankhurst. 1979. *The Interpretation of Igneous Rocks*, 450. George Allen and Unwin Boston.

Dilles, J. H., A. J. Kent, J. L. L. Wooden, et al. 2015. "Zircon Compositional Evidence for Sulfur -Degassing From or e -Forming Arc Magmas." *Economic Geology* 11 0 (1): 241–251.

Geopacific Resources Ltd. 2014. *Bonanza Gold Grades at Kou Sa, Cambodia: ASX Announcement*. Vol. 247, 13. Geopacific Resources. http://geopacific.com.au/index.php/asx-announcements/current-asx-announcements/.

Halpin, J. A., T. Jensen, P. McGoldrick, et al. 2014. "Authigenic Monazite and Detrital Zircon Dating From the Proterozoic Rocky Cape Group, Tasmania: Links to the Belt-Purcell Supergroup, North America." *Precambrian Research* 250: 50–67.

Horstwood, M. S. A., J. Košler, and G. Gehrels. 2016. "Community Derived Standards for LA ICP-MS U-(Th-) Pb Geochronology–Uncertainty Propagation, Age Interpretation and Data Reporting." *Geostandards and Geoanalytical Research* 40: 311–332.

10991034, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/gj.70352 by Cambodia Hinari NPL, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Irvine, T. N., and W. R. A. Baragar. 1971. "A Guide to the Chemical Classification of the Common Volcanic Rocks." *Canadian Journal of Earth Sciences* 8: 523–548.

Jacobson, H. S., C. T. Pierson, T. Japakasetr, et al. 1969. "Mineral Investigation in Northeastern Thailand." U.S.G.S, Professional Paper 618, 96.

Japan International Cooperation Agency. 2010. "Geological Map of Cambodia."

Jochum, K. P., and J. Enzweiler. 2014. "Reference Materials in Geochemical and Environmental Research." In *Treatise on Geochemistry*, vol. 15, 2nd ed., 43–70. Elsevier.

Kamvong, T., and K. Zaw. 2009. "The Origin and Evolution of Skarn-Forming Fluids From the Phu Lon Deposit, Northern Loei Fold Belt, Thailand: Evidence From Fluid Inclusion and Sulfur Isotope Studies." *Journal of Asian Earth Sciences* 34, no. 5: 624–633.

Kasahara, N., S. Niki, E. Uchida, K. Yarimizu, R. Cheng, and T. Hirata. 2021. "Zircon U–Pb Chronology on Plutonic Rocks From Northeastern Cambodia." *Heliyon* 7, no. 4: e06752.

Khositanont, S., K. Zaw, S. Meffre, Y. Panjasawatwong, P. Ounchanum, and T. Thanasuthipitak. 2013. "Geotectonic and Geochronology of Volcano-Plutonic Rocks in the Loei Petchabun Fold Belt." In *Proceedings of 2nd Lao Thai Technical Conference on Geology and Mineral Resources. Department of Geology and Minerals, Lao PDR*, 164–187. Department of Geology and Minerals, Laos.

Leslie, C., S. Meffre, D. R. Cooke, J. Thompson, N. Howard, and A. Barker. 2022. "Complex Petrogenesis of Porphyry-Related Magmas in the Cowal District, Australia: Insights From LA IC P-MS Zircon Imaging." In *Tectonomagmatic Influences on Metallogeny and Hydrothermal Ore Deposits*, 159–180. Society of Economic Geologists. https://doi.org/10.5382/sp.24.10.

Loader, M. A., J. J. Wilkinson, and R. N. Armstrong. 2017. "The Effect of Titanite Crystallization. On Eu and Ce Anomalies in Zircon and Its Implications for the Assessment of Porphyry Cu Deposit Fertility." *Earth and Planetary Science Letters* 472: 107–119. https://doi.org/10.1016/j.epsl.2017.05.010.

Loucks, R. R., M. L. Fiorentini, and G. J. Henriquez. 2020. "New Magmatic Oxybarometer Using Trace Elements in Zircon." *Journal of Petrology* 61. https://doi.org/10.1093/petrology/egaa034.

Lu, Y. J., R. R. Loucks, M. Fiorentini, et al. 2016. "Zircon Compositions as a Pathfinder for Porphyry Cu ± Mo ± Au Deposits." In *Tectonics and Metallogeny of the Tethyan Orogenic Belt*, edited by J. P. Richards. Society of Economic Geologists. https://doi.org/10.5382/SP.19.13.

Ly, P., S. Seang, K. Kret, et al. 2022. "Lithology, Hydrothermal Alteration, and Ore Characteristics of Area-1 in Koh Sla, Chhouk District, Kampot Province, Southern Cambodia." In *Proceedings of the International Symposium on Earth Science and Technology, Japan*. Department of Earth Sciences and Technology, Kyushu University.

Maniar, P. D., and P. M. Piccoli. 1989. "Tectonic Discrimination of Granitoid." *Geological Society of America Bulletin* 101: 635–643.

Martin, H. 1986. "Effect of Steeper Archean Geothermal Gradient on the Geochemistry of Subduction-Zone Magmas." *Geology* 14: 753–756.

Metcalfe, I. 2002. "Permian Tectonic Framework and Paleogeography of SE Asia." *Journal of Asian Earth Sciences* 20, no. 6: 551–566.

Metcalfe, I. 2011. "Tectonic Framework and Phanerozoic Evolution of Sundaland." *Gondwana Research* 19, no. 1: 3–21.

Metcalfe, I. 2013. "Gondwana Dispersion and Asian Accretion: Tectonic and Palaeogeographic Evolution of Eastern Tethys." *Journal of Asian Earth Sciences* 66: 1–33.

Muller, C. J. 1999. "Geochemistry, Fluid Characteristics and Evolution of the French Mine Gold 492 Skarn System, Eastern Thailand." Unpublished BSc (Hons) thesis, ARC Centre of 493 Excellence in Ore Deposits (CODES), University of Tasmania, Hobart, Australia, 145.

Nevolko, P. A., T. V. Svetlitskaya, A. A. Savichev, V. S. Vesnin, and P. A. Fominykh. 2021. "Uranium-Pb Zircon Ages, Whole-Rock and Zircon Mineral Geochemistry as Indicators for Magmatic Journal Pre-Proof Journal Pre-Proof 52 Fertility and Porphyry cu-Mo-Au Mineralization at the Bystrinsky and Shakhtama Deposits, Eastern Transbaikalia, Russia." *Oregon Geology* 139: 104532. https://doi.org/10.1016/j.oregeorev.2021.104532.

Nie, F., S. S. Liu, Y. F. Yang, Z. M. Peng, and L. N. Guo. 2019. "Zircon U-Pb Dating of the Diorite of Phu Lon Skarn Cu-Au Deposit in Thailand and Its Significance." *Sedimentary Geology and Tethyan Geology* 39: 71–78.

Ouyang, Y., H. Liu, F. Nie, et al. 2019. "Petrogenesis of Middle Triassic Andesite in Sayaburi Area, Laos: Constraints From Whole-Rock Geochemistry, Zircon U-Pb Geochronology, and Sr-Nd Isotopes." *Journal of Central South University* 26: 3502–3515.

Pearce, J. A., N. B. Harris, and A. G. Tindle. 1984. "Trace Element Discrimination Diagrams for the Tectonic Interpretations of Granitic Rocks." *Journal of Petrology* 25, no. 4: 956–983. https://doi.org/10.1093/petrology/25.4.956.

Pizarro, H., E. Campos, F. Bouzari, et al. 2020. "Porphyry Indicator Zircons (PIZs): Application to Exploration of Porphyry Copper Deposits." *Ore Geology Reviews* 126: 103771. https://doi.org/10.1016/j.oregeorev.2020.103771.

Qian, X., Q. Feng, W. Yang, Y. Wang, C. Chonglakmani, and D. Monjai. 2015. "Arc-Like Volcanic Rocks in NW Laos: Geochronological and Geochemical Constraints and Their Tectonic Implications." *Journal of Asian Earth Sciences* 98: 342–357.

Richardson, C. T., J. R. Craig, M. Adrian, and D. Kurtis. 2016. "Controls on Polymetallic Vein Deposits and Porphyry Deposits in the Phum Syarung-Dok Yong Fault Corridor, Ratanakiri Province, Cambodia." Unpublished Poster, Society of Economic Geologists, 2016.

Rodmanee, T. 2000. "Genetic Model of Phu Thab Fah Gold Deposit Ban Huai Phuk Amphoe Wang Saphung Changwat Loei." Doctoral diss., Chiang Mai: Graduate School, Chiang Mai University, 2000.

Salam, A., K. Zaw, S. Meffre, J. McPhie, and C. K. Lai. 2014. "Geochemistry and Geochronology of Epithermal au-Hosted Chatree Volcanic Sequence: Implication for the Tectonic Setting of the Loei Fold Belt in Central Thailand." *Gondwana Research* 26, no. 1: 198–217.

Samnang Angkor Development. 2019. "Semi-Annual Exploration Report."

Seang, S., K. Yonezu, T. Tindell, T. Nakanishi, K. Watanabe, and J. Pelletier. 2019. "Lithogeochemistry of Intrusive Rocks in the Halo Porphyry Copper-Molybdenum Prospect, Northeast Cambodia." *Open Journal of Geology* 9, no. 7: 342–363.

Sláma, J., J. Košler, D. J. Condon, et al. 2008. "Plešovice Zircon—A New Natural Reference Material for U–Pb and Hf Isotopic Microanalysis." *Chemical Geology* 249, no. 1–2: 1–35.

Sone, M., and I. Metcalfe. 2008. "Parallel Tethyan Sutures in Mainland Southeast Asia: New Insights for Palaeo-Tethys Closure and Implications for the Indosinian Orogeny." *Comptes Rendus Geoscience* 340, no. 2–3: 166–179.

Stacey, J. S., and J. D. Kramers. 1975. "Approximation of Terrestrial Lead Isotope Evolution by a Two-Stage Model." *Earth and Planetary Science Letters* 26: 207–221. https://doi.org/10.1016/0012-821X(75)90088-6.

Sun, S. S., and W. F. McDonough. 1989. "Chemical and Isotopic Systematic of Oceanic Basalts: Implication for Mantle Composition and Processes." In *Magmatism in Ocean Basins*, edited by A. D. Saunder and M. J. Norry, 313–345. Geological Society, Special Publication.

10991034, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/gj.70352 by Cambodia Hinari NPL, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Thassanapak, H., M. Udchachon, C. Burrett, and Q. Feng. 2017. "Geochemistry of Radiolarian Cherts From a Late Devonian Continental Margin Basin, Loei Fold Belt, Indo-China Terrane." *Journal of Earth Science* 28: 29–50.

Thompson, J. M., S. Meffre, and L. Danyushevsky. 2018. "Impact of Air, Laser Pulse Width, and Fluence on U–Pb Dating of Zircons by LA-ICPMS." *Journal of Analytical Atomic Spectrometry* 33: 221–230.

Uchida, E., S. Nagano, S. Niki, et al. 2023. "U Pb Dating for Zircons From Granitic Rocks in Southwestern Cambodia." *Heliyon* 9: e19734. https://doi.org/10.1016/j.heliyon.2023.e19734.

Udchachon, M., H. Thassanapak, Q. Feng, and C. Chonglakmani. 2011. "Geochemical Constraints on the Depositional Environment of Upper Devonian Radiolarian Cherts From Loei, North-Eastern Thailand." *Frontiers of Earth Science* 5: 178–190.

Ueno, K., and T. Charoentitirat. 2011. "Carboniferous and Permian." In *The Geology of Thailand*, edited by M. F. Ridd, A. J. Barber, and M. J. Crow, 71–136. Geological Society of London.

Wiedenbeck, M. A. P. C., P. Alle, F. Y. Corfu, et al. 1995. "Three Natural Zircon Standards for U- Th- Pb, Lu- Hf, Trace Element and REE Analyses." *Geostandards Newsletter* 19, no. 1: 1–23.

Zaw, K., S. Meffre, C. K. Lai, et al. 2014. "Tectonics and Metallogeny of Mainland Southeast Asia Review and Contribution." *Gondwana Research* 26, no. 1: 5–30.

Zhang, Y. Z., Y. J. Wang, B. Srithai, and B. Phajuy. 2016. "Petrogenesis for the Chiang Dao Permian High-Iron Basalt and Its Implication on the Paleo-Tethyan Ocean in NW Thailand." *Journal of Earth Science* 27, no. 3: 425–434.

Zhang, Y. Z., X. Yang, Y. J. Wang, et al. 2021. "Rifting and Subduction Records of the Paleo-Tethys in North Laos: Constraints From Late Paleozoic Mafic and Plagiogranitic Magmatism Along the Song Ma Tectonic Zone." *GSA Bulletin* 132, no. 1–2: 212–232.

Zhang, Y. Z., X. Q. Yu, Y. J. Wang, et al. 2023. "Reconstructing the East Palaeo-Tethyan Assemblage Boundary in West Indonesia: Constraints From Triassic Granitoids in the Bangka and Belitung Islands." *Geological Society, London, Special Publications* 531: 265–286.

Zou, X. Y., K. Z. Qin, X. L. Han, et al. 2019. "Insights Into Zircon RE E Oxy-Barometers: A Lattice Strain Model Perspective." *Earth and Planetary Science Letters* 506: 87–96. https://doi.org/10.1016/j.epsl.2018.10.031.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** U-Pb age data obtained from zircon from the diorite and granite from Phnom Sro Ngam and the granodiorite and quartz-feldspar porphyry from the Halo Prospect.

10991034, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/gj.70352 by Cambodia Hinari NPL, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License